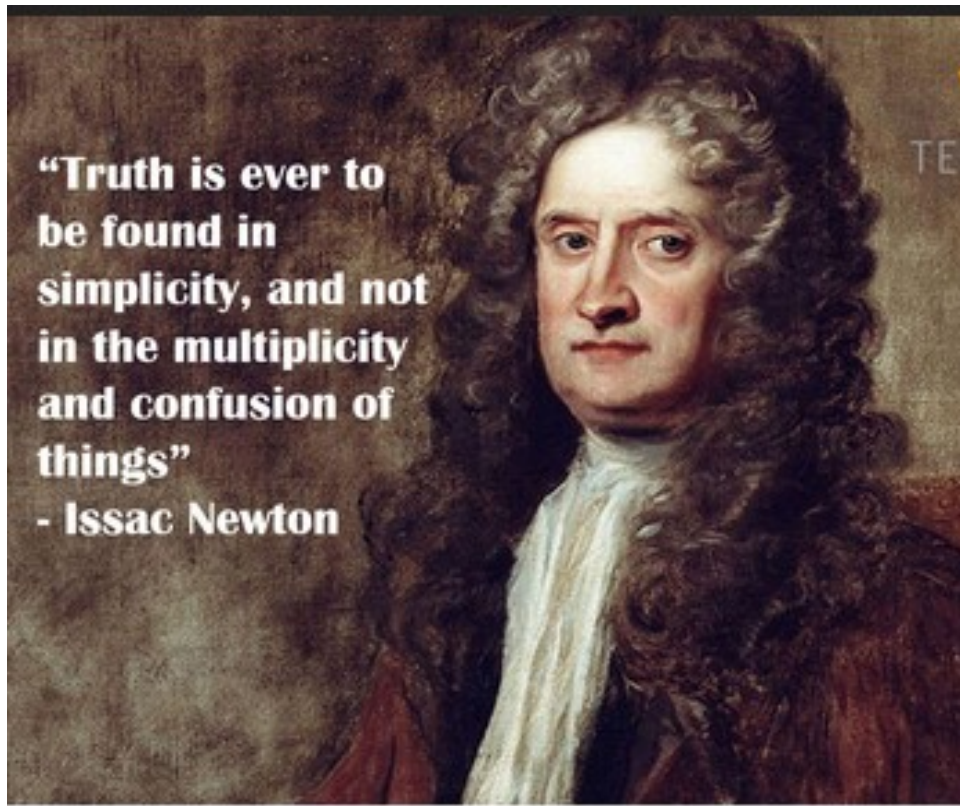


Newton's laws – slide 1
impulse-momentum theorem – slide 26
Newton's second law in 2D – slide 55
gravitational law – slide 71
weightlessness



Newton's first law :law of inertia

- **Newton's First Law of Motion** states:
An object will remain at rest or in motion with constant velocity unless acted on by a net external force.
- An **external force** is a force applied to the object from some other object.
 - force from an impact, gravity, air resistance, etc
 - Net force = 0 means acceleration = 0
Velocity = constant
same magnitude and same direction

Inertia is the natural tendency of an object to remain at rest in motion at a constant speed along a straight line.

The *mass* of an object is a quantitative measure of inertia.

SI Unit of Mass: kilogram (kg)

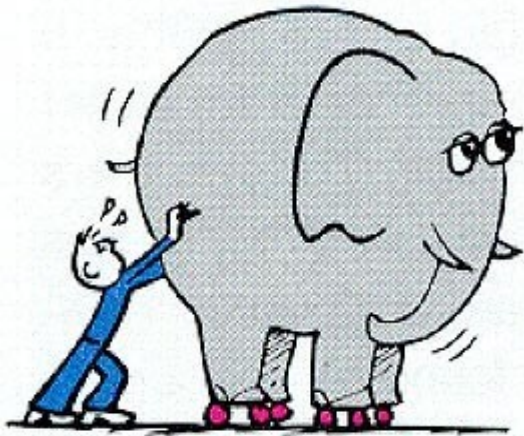
misconception: " **Motion requires force** " **WRONG**.

A puck initially hit will keep moving on the ice until it is stopped by a wall or by a player.

Motion does not require force.

A net Force is only required to change the motion.

to speed up an object, to slow it down, to stop it , to get it to move or to change its direction,



Mass is amount of “ stuff” or matter in an object. Mass is a measure of inertia. How much an object resists a change in motion. Its laziness to change Its direction or speed.

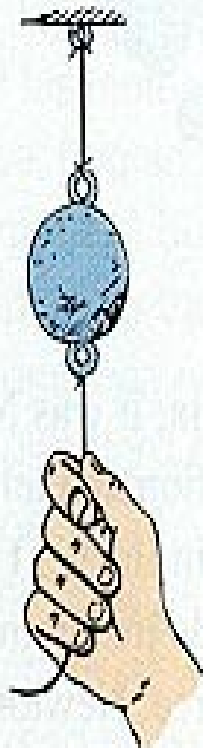
Source: Conceptual Physics, Paul Hewitt

Demo : inertia

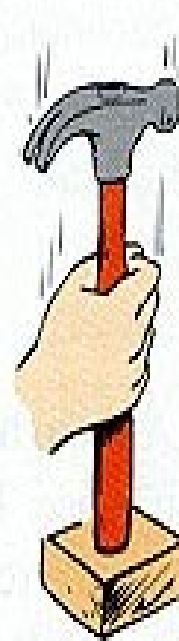
Videos: crash test / inertia loop / inertia string / table cloth 1 & 2 /



Why will the coin drop into the glass when a force accelerates the card?



Why is it that a slow continuous increase in the downward force breaks the string above the massive ball, but a sudden increase breaks the lower string?



Why does the downward motion and sudden stop of the hammer tighten the hammerhead?



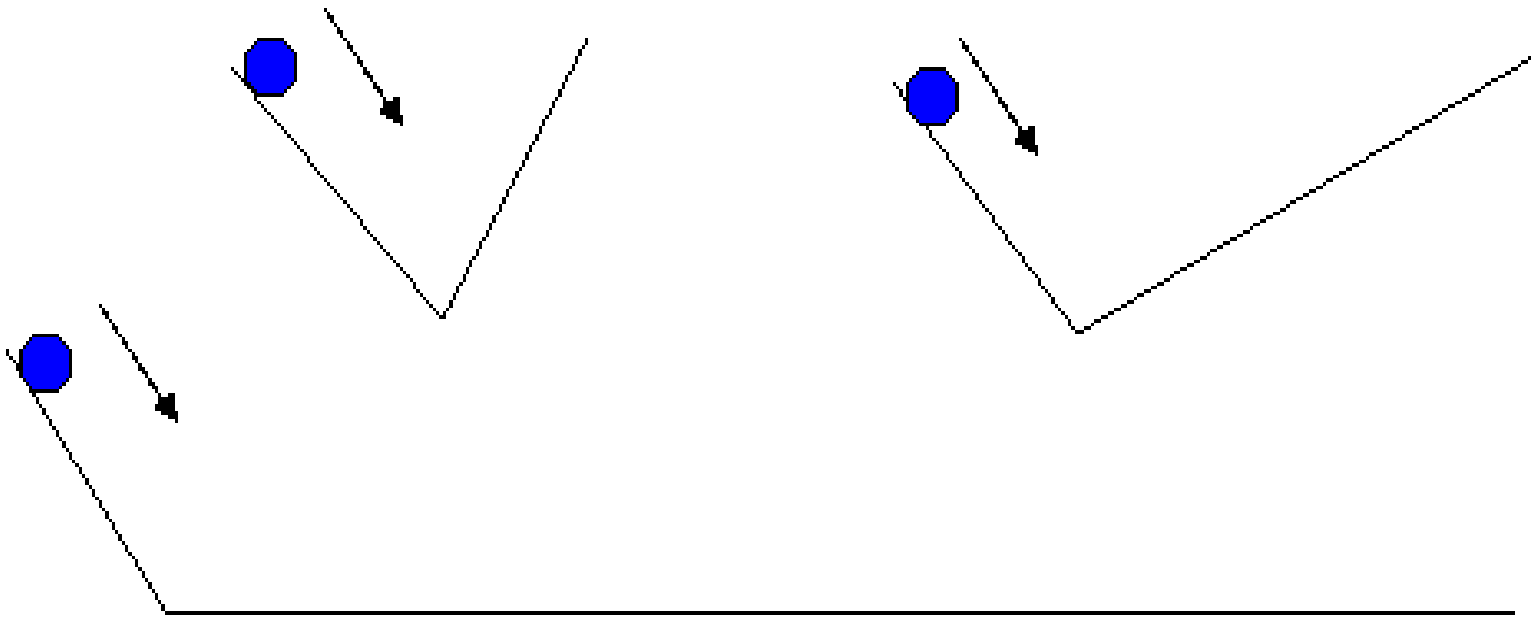
**WHAT HAPPEN TO THE PASSENGERS
OF A CAR WHEN IT ABRUPTLY BREAKS ?
OR SPEED UP ?
WHY DO YOU NEED TO WEAR A SEAT BELT ?**



**HOW CAN YOU EXPLAIN THE TABLE TRICK
USING INERTIA PRINCIPLE ?**

USING A THOUGHT EXPERIMENT, GALILEO REASONED THAT ARISTOTLE WAS WRONG.

IF THERE IS NO NET FORCE, AN OBJECT IN MOTION STAYS IN MOTION, AT THE SAME SPEED AND THE SAME DIRECTION. Describe this experiment:



An elephant has a large inertia and can run very fast. What could you do to escape a charging elephant ? (using Newton;'s 1st law)



stinger missiles are heat seeking missiles. They detect infra red and can hit planes. They reach them because the " feel " the heat. If you are an experienced pilot, how can you avoid the stingers ?



**Let s say you try to push a large mass like an Anvil.
Is it harder to push it on Earth or in space, out of gravity reach ? (tricky)**



A pipe is bent into the shape shown and oriented so that it is sitting horizontally on a table top. You are looking at the pipe from above. The interior of the pipe is smooth. A marble is shot into one end and exits the other end. Which one of the paths shown in the drawing will the marble follow when it leaves the pipe?

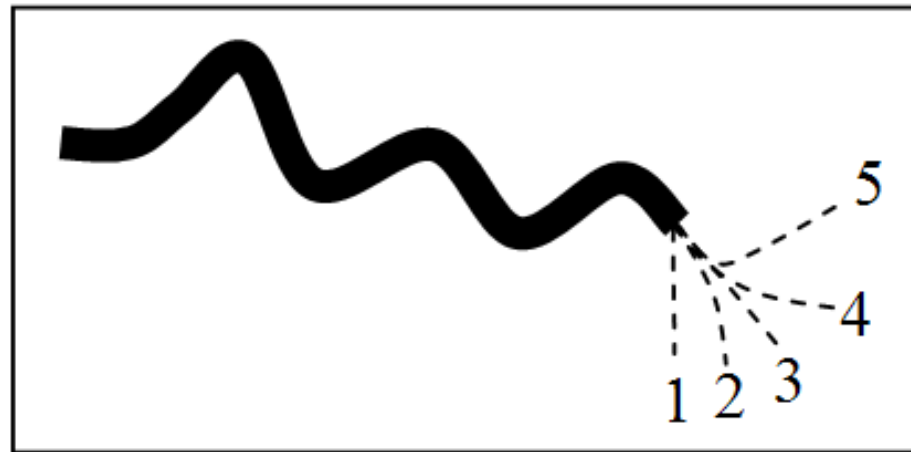
a) 1

b) 2

c) 3

d) 4

e) 5



See video clip from “wanted” → Not Physics.

Which one of the following terms is used to indicate the natural tendency of an object to remain at rest or in motion at a constant speed along a straight line?

- a) velocity
- b) equilibrium
- c) acceleration
- d) inertia
- e) force

DADDY'S HOME

10. What is the purpose of the study?

I SAY! THAT FALLING
APPLE HAS MADE
ME REALIZE SOMETHING...

I'M HUNGRY!

NEWTON'S FIRST LAW OF MOTION

An object in motion tends to stay in motion.

An object at rest tends to stay at rest...

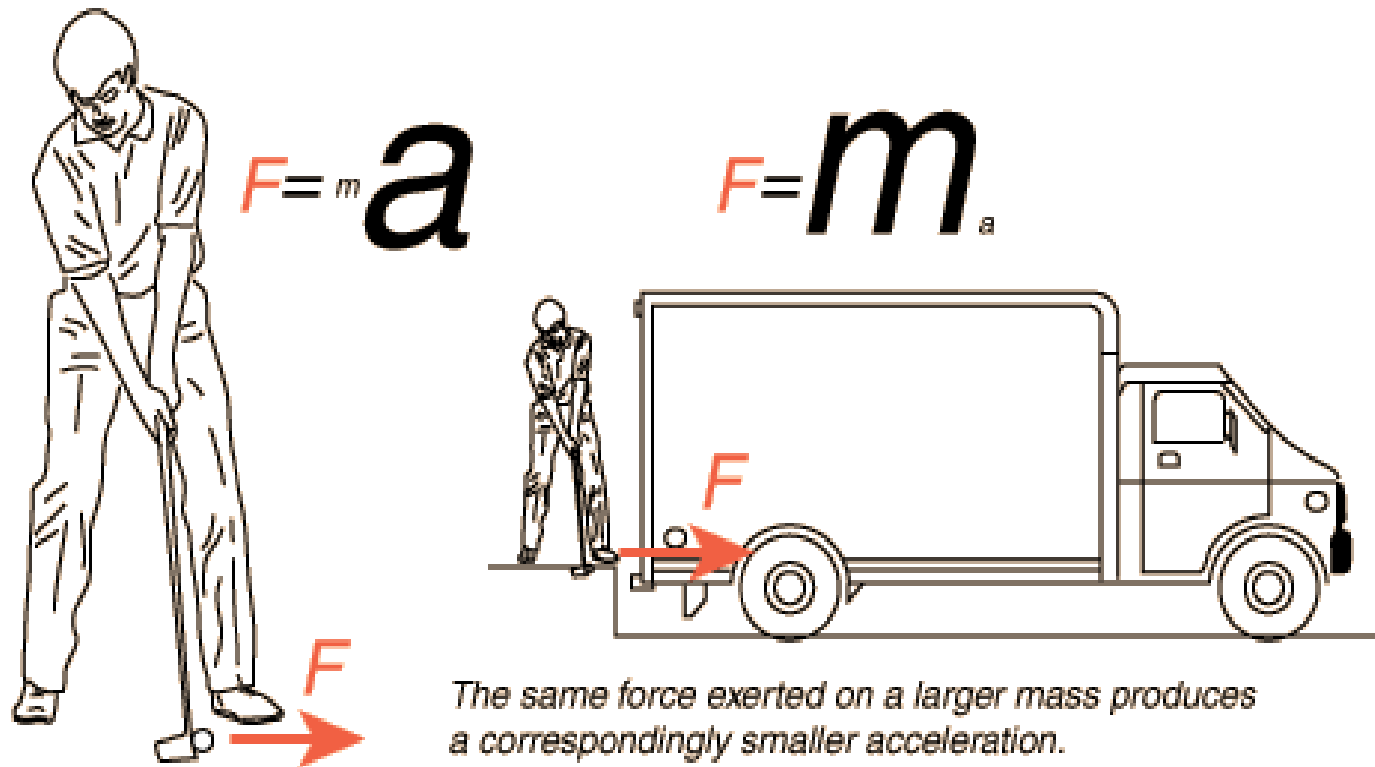
unless acted upon by an object in motion.

NEED HELP, NOW?

NO, SIT!
I'VE GOT IT!

Newton's second law : $F = ma$

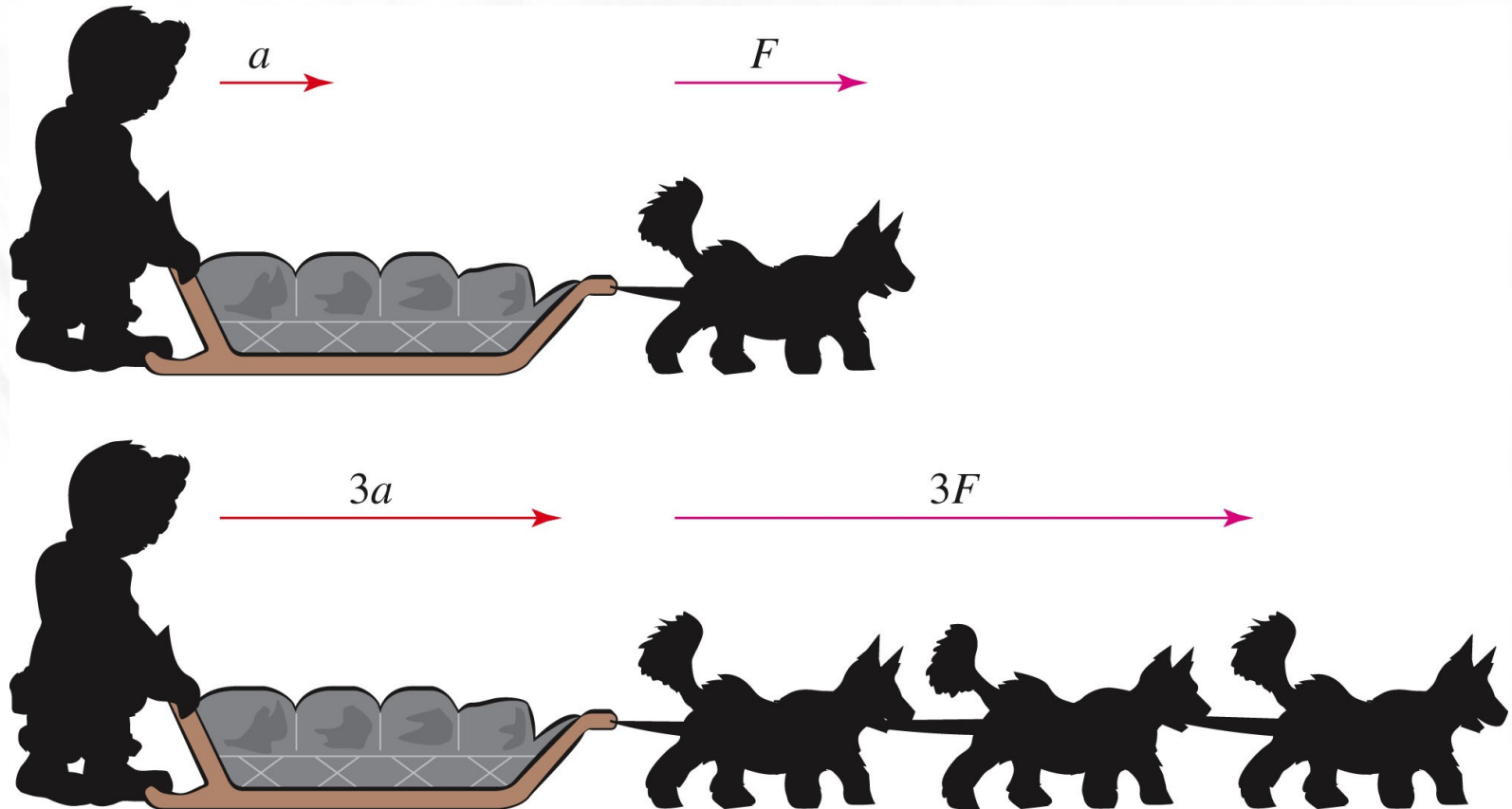
Push/pull = change in velocity (magnitude or direction or both)
but the change depends on the mass !!!
Force in newtons, mass in kg, acceleration in m/s/s



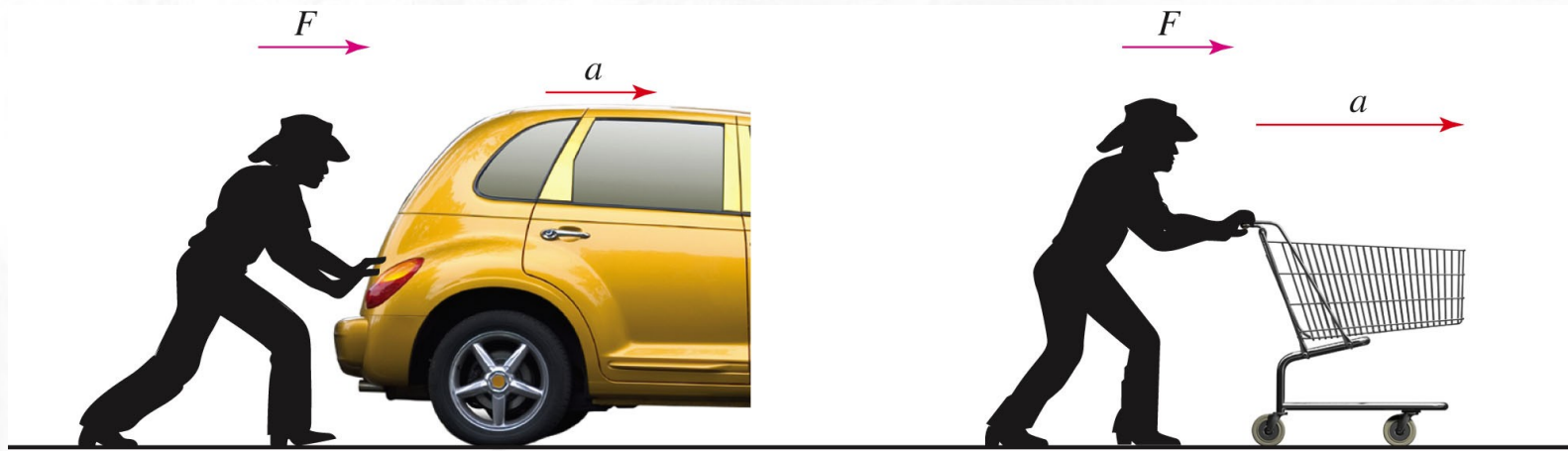
Exploration of physical sciences Newton's law

Sims <https://phet.colorado.edu/en/simulation/forces-and-motion-basics>

Net Force = mass x acceleration
(Newtons) (kg) (m/s/s)



For the same mass , the acceleration is proportional to the net force



For the same force, the acceleration is inversely proportional to the mass. Acceleration = $F_{\text{net}}/\text{mass}$

At take off, both masses (baby, adult) will get to the same acceleration. Adults have more mass
 → more push from the chair. They feel more
 Pushing them

If you are 200 pounds in free-fall → 1 g
 But you are pushed with 200pounds on ice → 1g
 Your own weight is pushing you.
 If 2g → twice your weight pushing you.
 Space shuttle 3gs → 3 times your weight pushing.
 Pilot → 10gs max. Pushing them so get to the right a

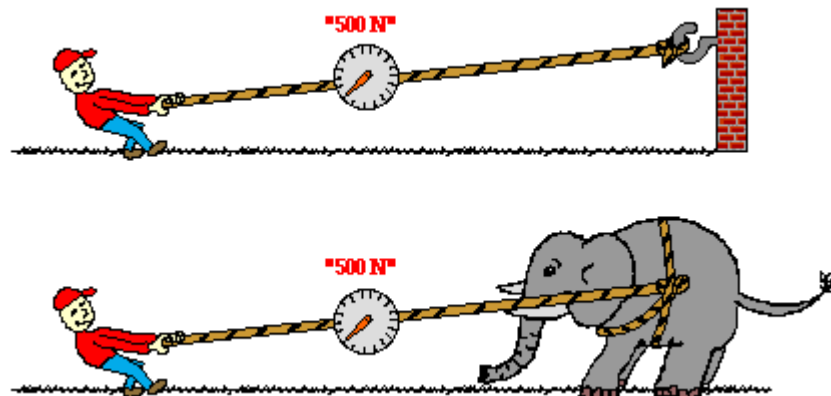


Complete the following statement: The term net force most accurately describes

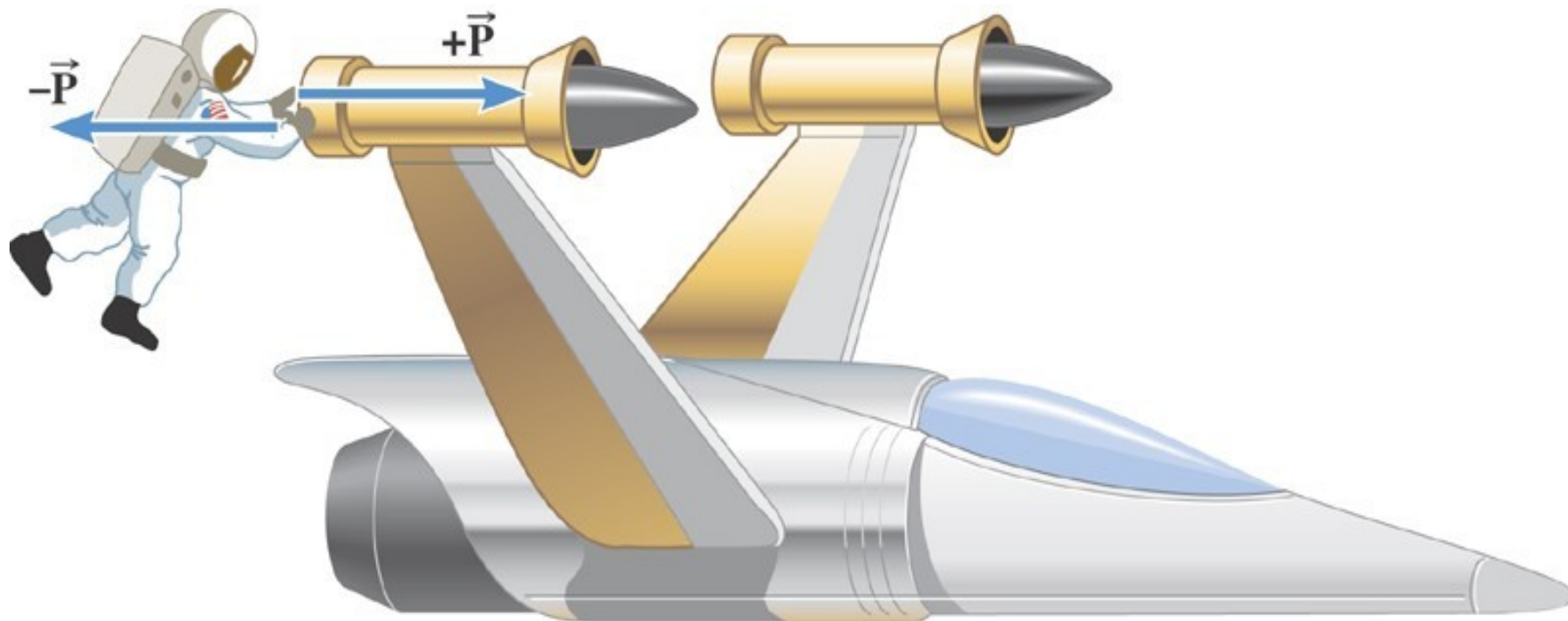
- a) the inertia of an object.
- b) the quantity that causes displacement.
- c) the quantity that keeps an object moving.
- d) the mass of an object.
- e) the quantity that changes the velocity of an object.

Newton's Third Law of Motion

Whenever one body exerts a force on a second body, the second body exerts an oppositely directed force of equal magnitude on the first body.



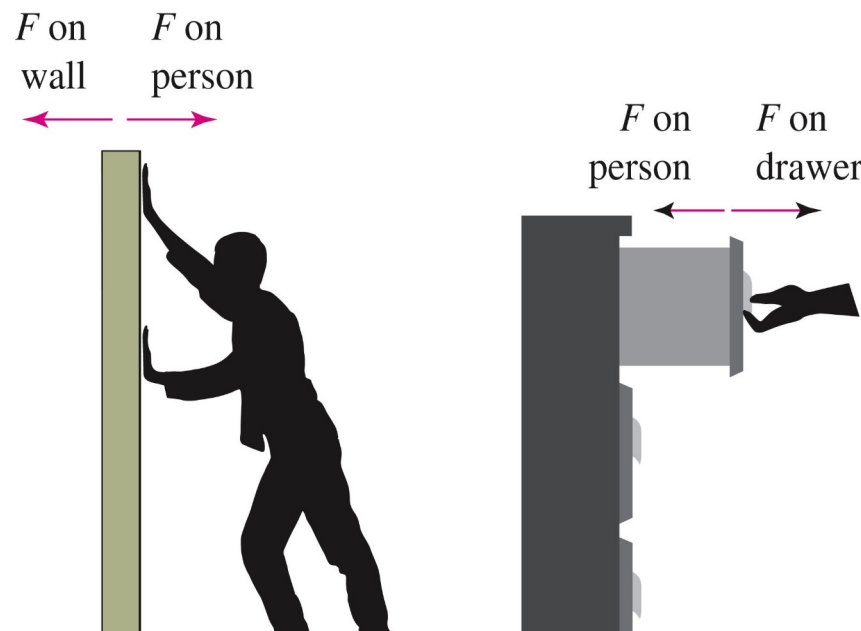
Demo with the scales



$$m \mathbf{a} = \text{Push} = \mathbf{M} \mathbf{a}$$

Newton's third law of motion

- **Newton's 3rd Law of Motion** states that forces always occur in pairs:
 - When one object exerts a force on a second object, the second exerts an equal and opposite force on the first.



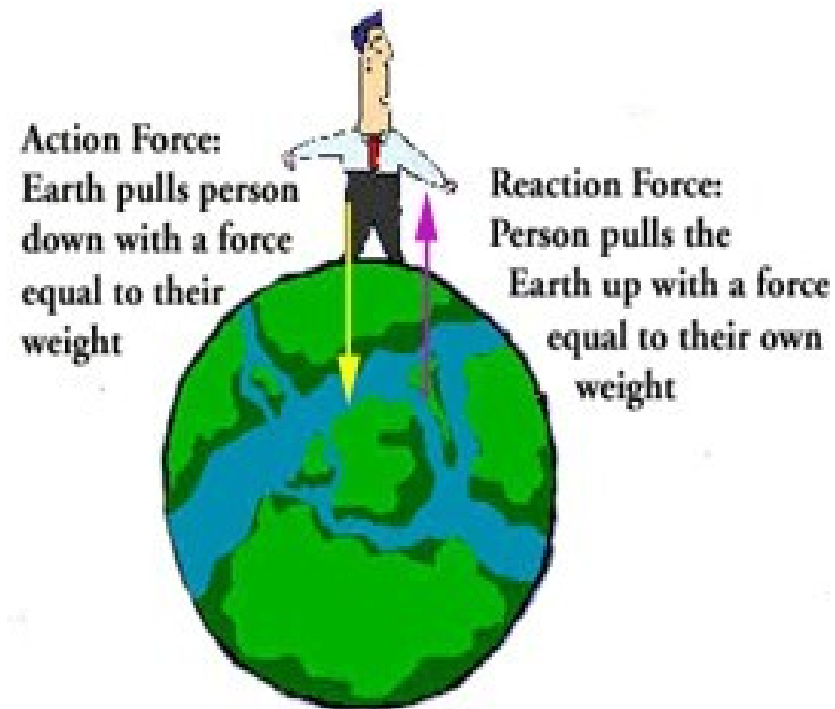
Newton's third law of motion,

cont'd

- If object A exerts a force on object B, then object B exerts an equal force in the opposite direction on A:

- $F_{A \text{ on } B} = - F_{B \text{ on } A}$

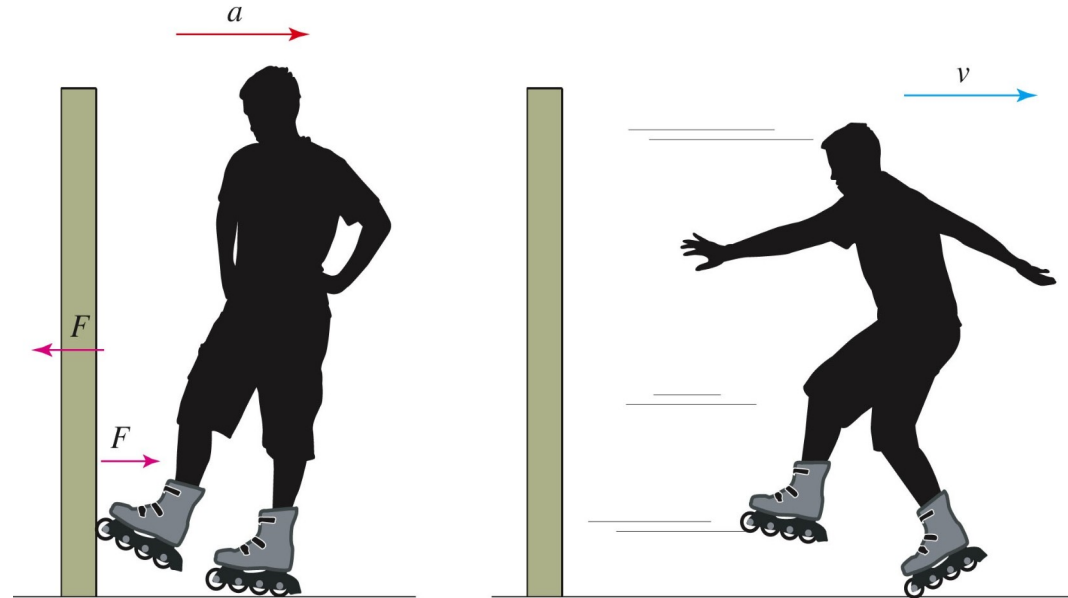
When you fall down, you feel the Earth exerting a force on you but you also exert that same force on the Earth.



Newton's third law of motion,

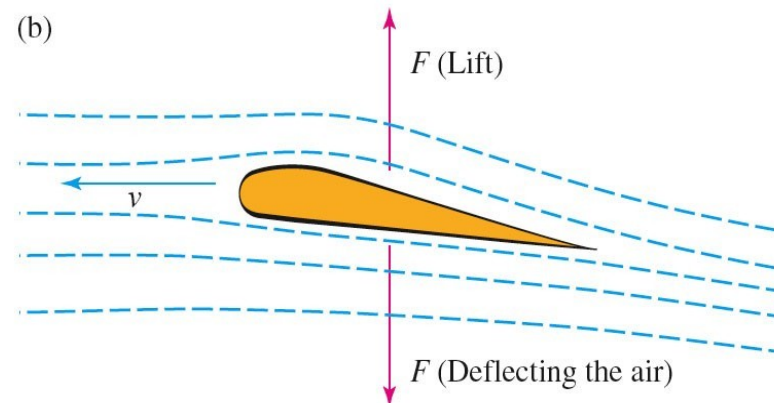
cont'd

- Think about pushing off against a wall.
 - You push against the wall.
 - The wall pushes back.
 - If the wall is weak, it might fall down.
 - If not, you move away.



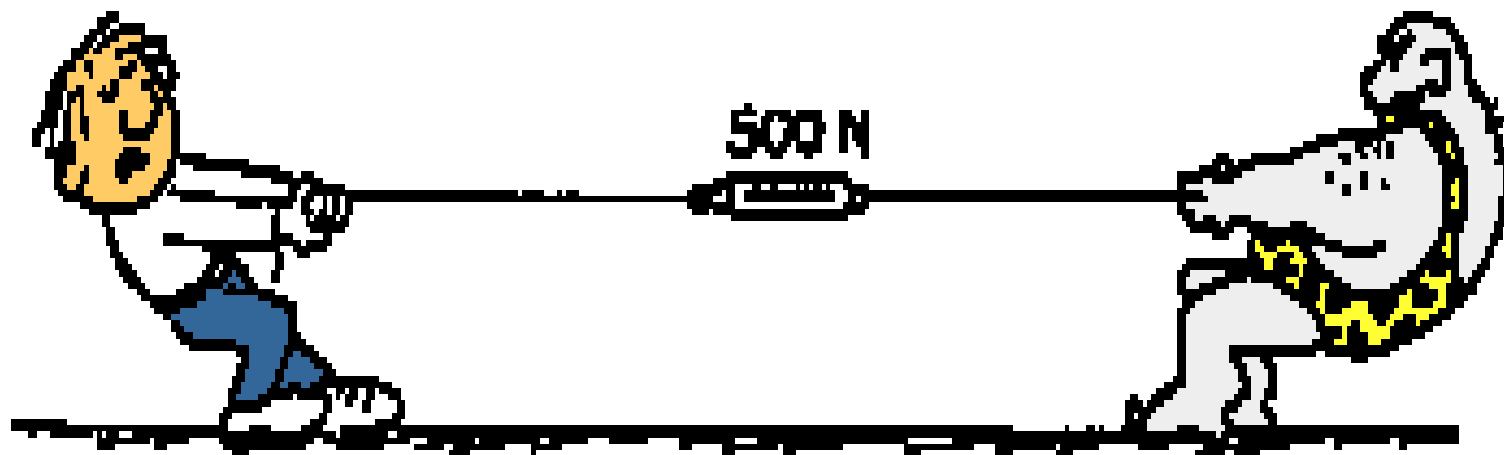
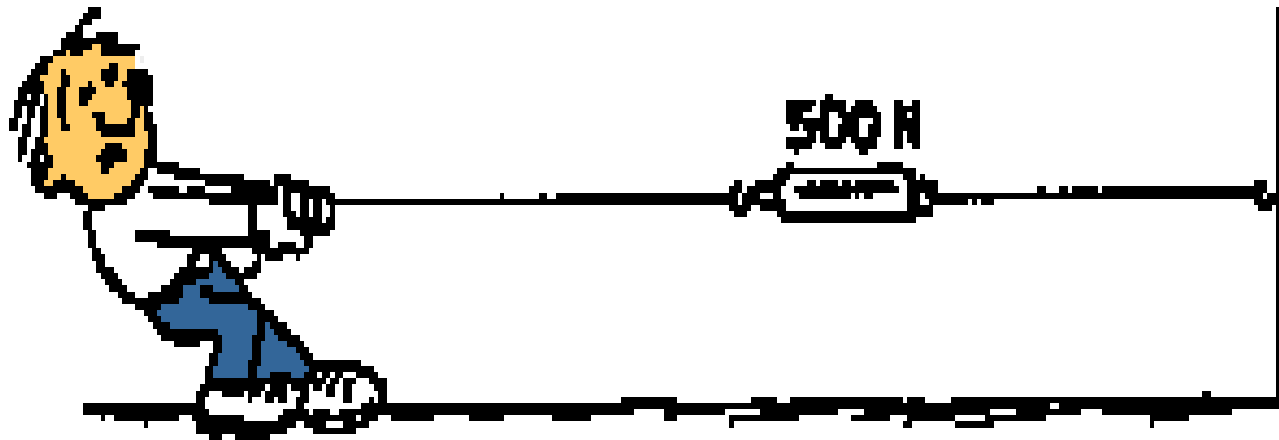
Newton's third law of motion, cont'd

- Consider an airplane's wing.
 - Due to the angle of attack, the air impacts the bottom of the wing.
 - The wing pushes the air out of the way.
 - The air pushes back and provides some lift.



https://www.reddit.com/r/wallstreetbets/comments/pv4xkk/a_summary_of_the_last_week_for/

NEWTON's THIRD LAW

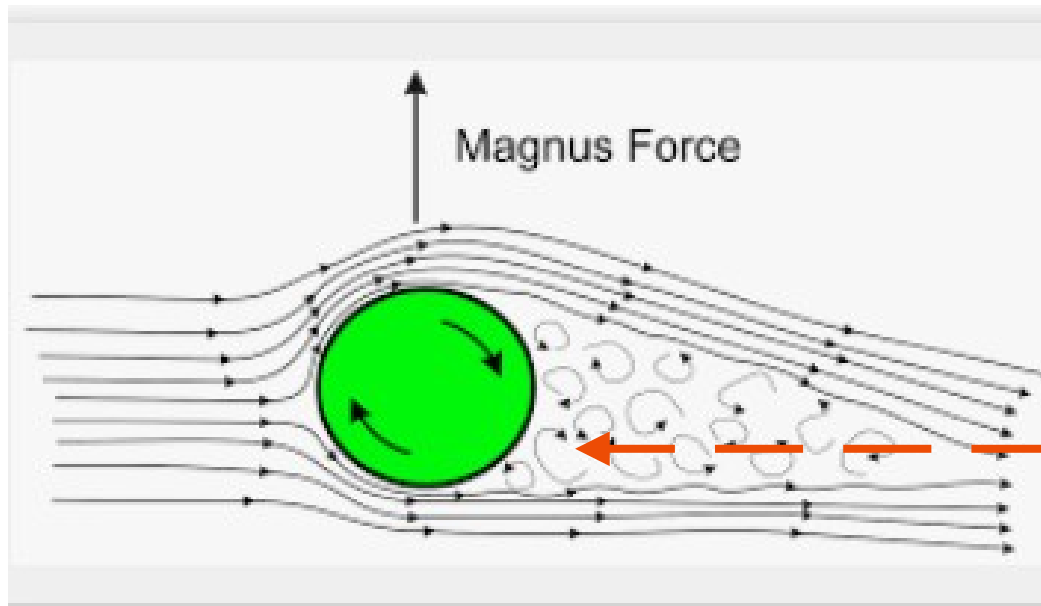
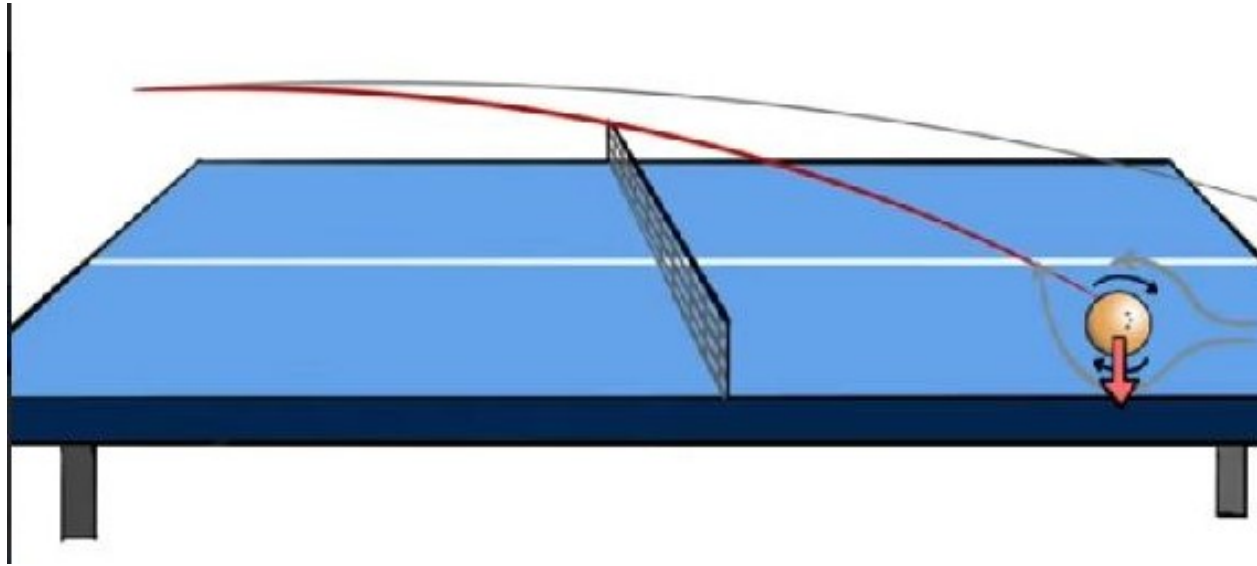


$-m A$

$=F =$

$M a$

Magnus force → consequence of Newton's third law.



See video magnus

- 1) a truck and a smart collide head front.
A) they both experience the same force
B) the truck experiences more force
C) the smart experiences more force
D) not enough information

What about the acceleration and damage ? (change in velocity)

- 2) A little girl is playing tug of war with her dad.
A) she exerts the greatest force
B) he exerts the greatest force
C) They experience the same pull

Is the acceleration the same ? (who experiences a change in speed)

3) An apple is placed on a table. The Earth pulls on the apple. What is the reaction ? Consequence?

4) A hammer hits a nail. What is the reaction ? Consequence?

5) The blades of a helicopter push the air down. What is the reaction. Consequence ?

6) while you walk you push the ground back ward (possible if friction)
What is the reaction ? Consequence?

$$m \mathbf{a} = \text{same force} = M \mathbf{a}$$

How Newton's third law can explain
The lift of the space shuttle ?

What happen to your shoulder if you shoot using
a riffle ? To spare your shoulder, is it better to
use a heavy riffle or a light one?

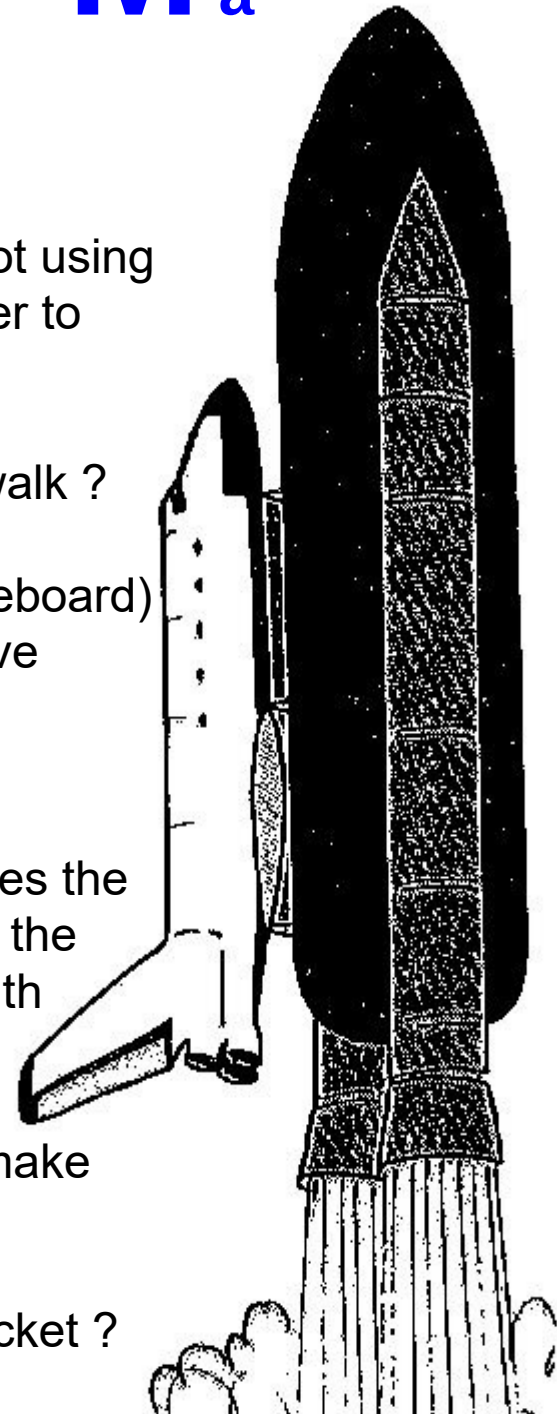
How Newton's third law can explain the walk ?

If you stand on the ground (not on a skateboard)
and you push the wall, why don't you move
back ? What is the other pair of forces
involved ?

In a cowboy movie, the good guy punches the
bad guy who goes flying away. Of course the
good guy stays steady. What is wrong with
that ?

Also, when a bullet hit a person, does it make
sense to see the guy flying off ?

How to relate newton's third law to the rocket ?



DURING A COLLISION THE FORCE and THE TIME
 “felt” BY THE 2 OBJECTS is the same BUT NOT THE
 ACCELERATION (or CHANGE OF SPEED= DAMAGE).



They both feel the same force
 But Baloo has a larger mass He does not
 move as much as Mowgli. Change in speed
 Is larger for Mowgli.

$$m \mathbf{a} = \mathbf{F} = - M \mathbf{a}$$

Same because
 Acceleration = change of speed over
 time. Time of collision is the same

Same force →

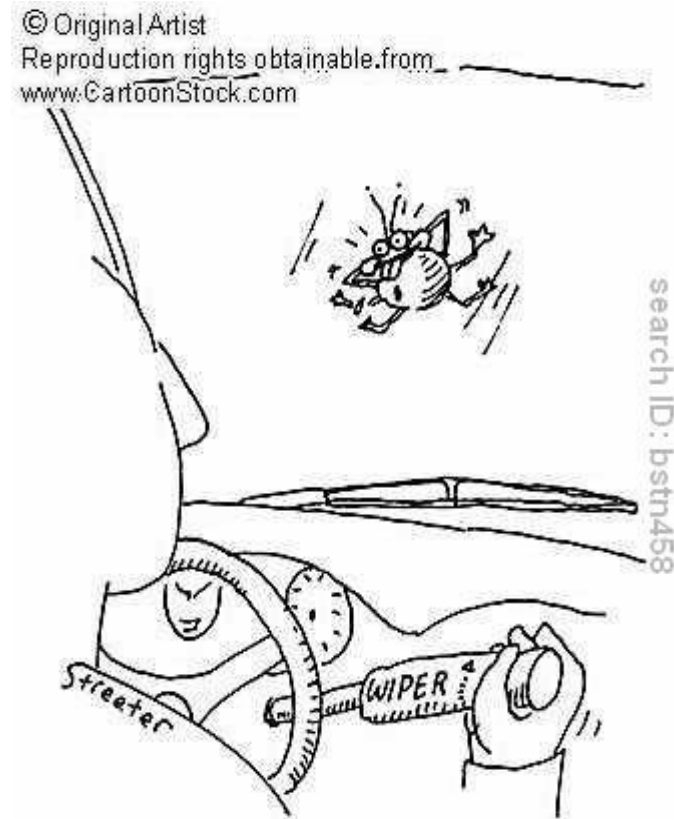
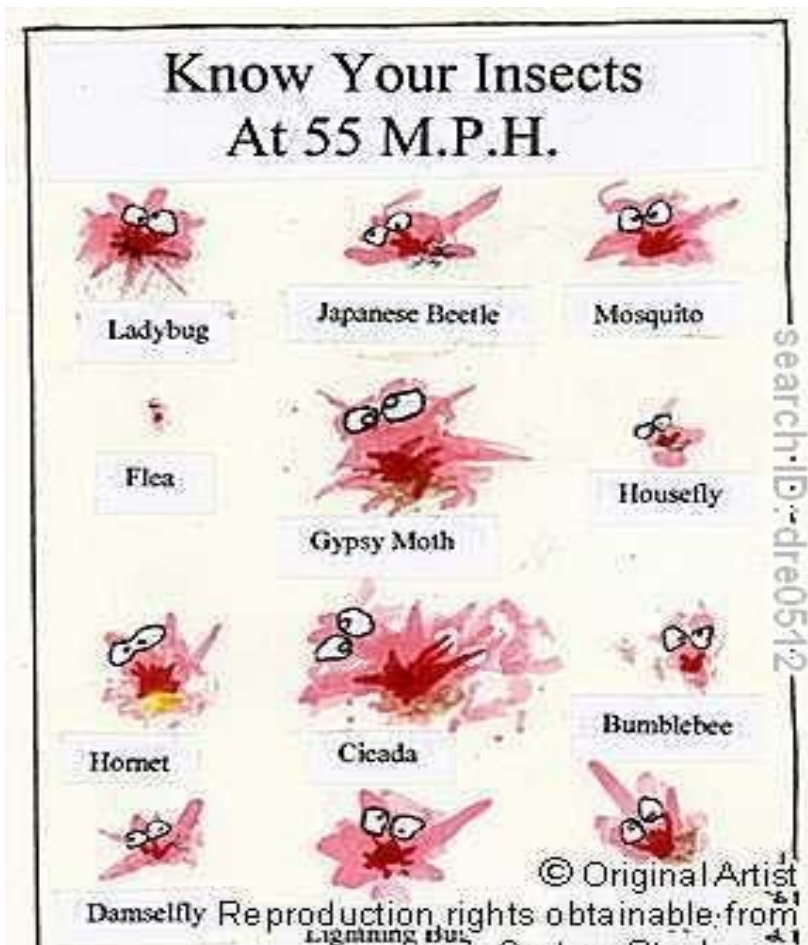
BIG MASS = small change in speed
 Small mass = BIG CHANGE IN SPEED

$$m \Delta \mathbf{V} = \mathbf{F} \times \text{time} = - M \Delta \mathbf{V}$$

You are traveling in a bus at highway speed on a nice summer day and an unlucky bug splatters on the front window.

With Newton's third law in mind , Compare to the force that acts of the bug , how much force acts of the bus ?

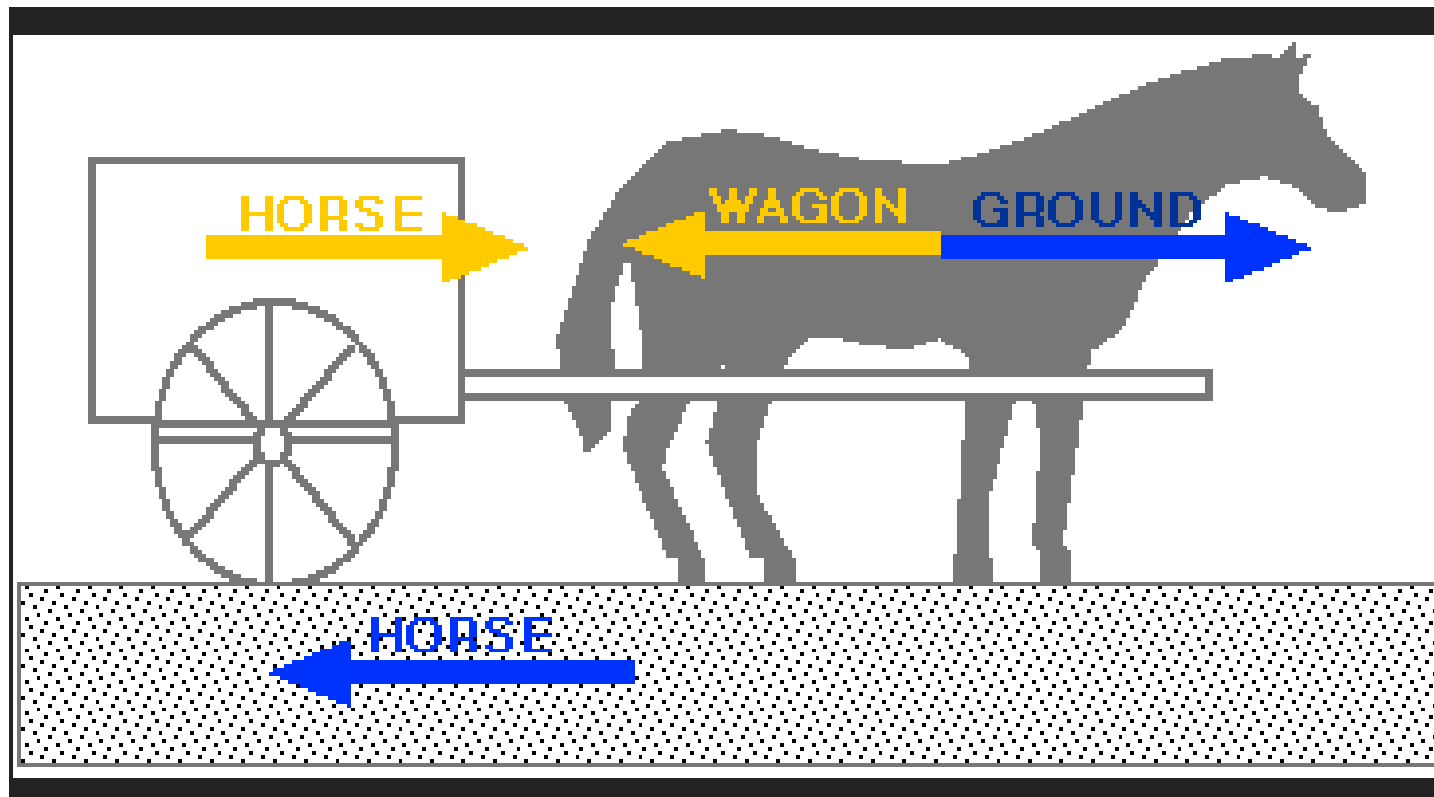
Which undergoes the greater acceleration ? Which therefore suffer the greater damage ?



A cell phone is sitting on a desk. Which one of the following is the reaction force to the cell phone's weight on the desk?

- a) the gravitational force on the cell phone
- b) the gravitational force on the table
- c) the normal force of the Earth on the table
- d) the normal force of the cell phone on the table
- e) the normal force of the table on the cell phone

Is there a paradox ?

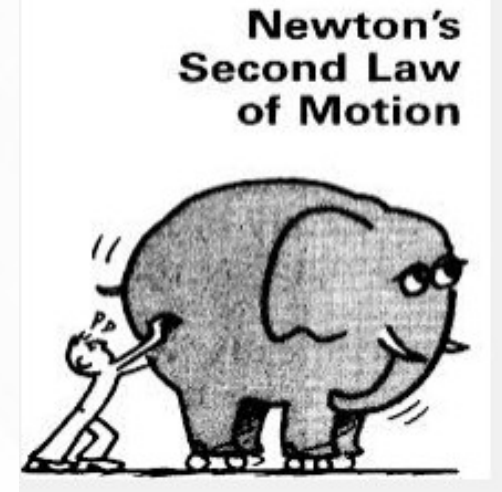


Newton's second law:

1) $F=ma$

2) impulse J

2) Newton's law in parts in 2D



Newton's second law of motion can be formally stated as follows: The acceleration of an object as produced by a net force is directly proportional to the magnitude of the net force, in the same direction as the net force, and inversely proportional to the mass of the object



Newton's second law: different approach. Let's write the law the same way Newton wrote it in his Principia.

BIG IDEA:

**To change the motion of an object, a force has to be applied
During a time t . Change in motion means change in velocity
(direction or magnitude).**

**Try to translate that statement in a mathematical relationship.
(force F during time t means change in velocity ΔV)**

So $\Delta V =$

.

But wait !

the change in velocity is inversely proportional to the mass (inertia).

**So, modify the relationship so the change in velocity is inversely proportional
To the mass (inertia)**

BIG IDEA: (Newton's second law)

If you apply a force **F** during a time **t**, the state of motion
Changes, The velocity increases by ΔV

Like wise, if an objects undergoes a change in motion
that's because it is acted upon by a force during a time

$$\mathbf{F} \Delta t = m \Delta V$$

$$\mathbf{J} = \mathbf{F} \Delta t$$

Impulse = change in momentum

The quantity $\mathbf{F} \Delta t$
is called the impulse and is noted
J in your book.

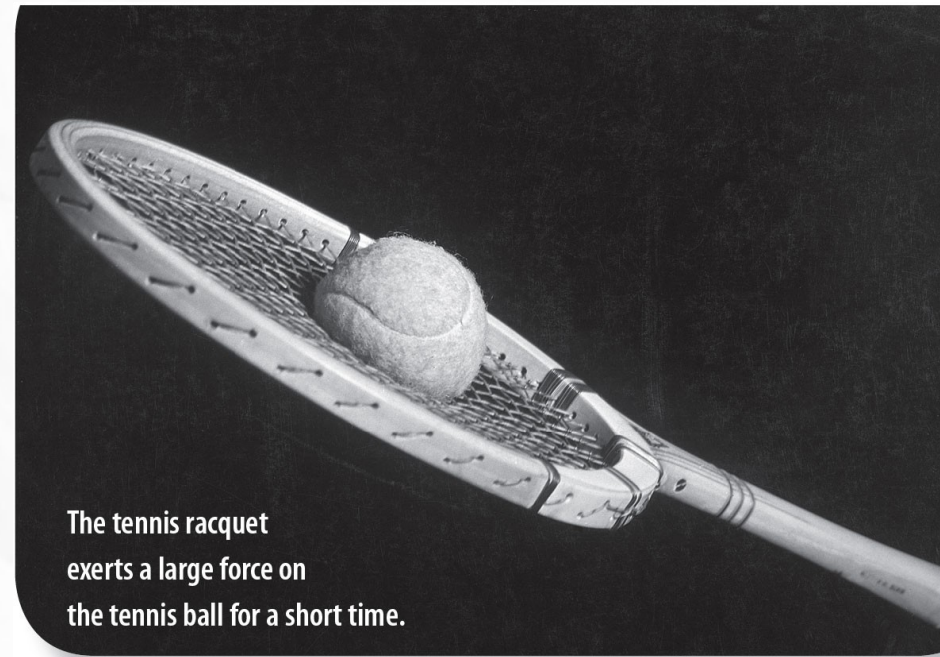
It is a vector because **F** is a vector. Units are N.s

The quantity $m \Delta V$ quantifies the
change of motion. It is called the change
In momentum.

It is a vector because **F** is a vector. Units are kg. m/s ³²

Example

Let's estimate the average force on a tennis ball as it is served. The ball's mass is 0.06 kg and it leaves the racket with a speed of 40 m/s. High-speed photography indicates that the contact time is about 5 milliseconds.



So the velocity of the ball increases from 0 to 40 because
There is a force applied to it during a time 0.005s

$$F\Delta t = m\Delta V$$

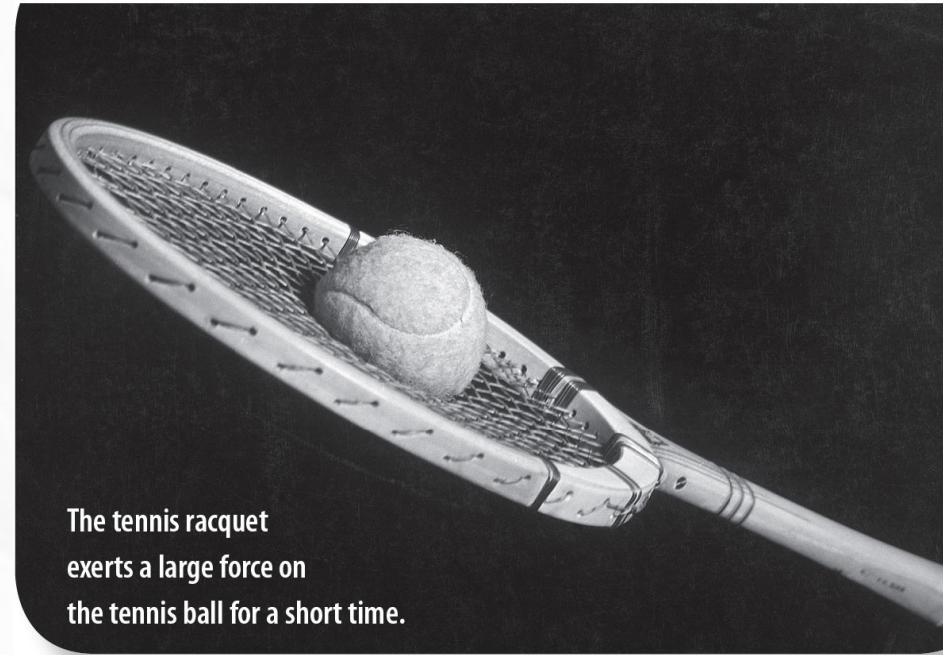
$$F\Delta t = m\Delta V$$

Let's estimate the average force on a tennis ball as it is served. The ball's mass is 0.06 kg and it leaves the racquet with a speed of 40 m/s. High-speed photography indicates that the contact time is about 5 milliseconds.

$$0.06(40-0) = F(0.005)$$

$$F = 480\text{N}$$

(about the weight of 48kg or 96pounds)

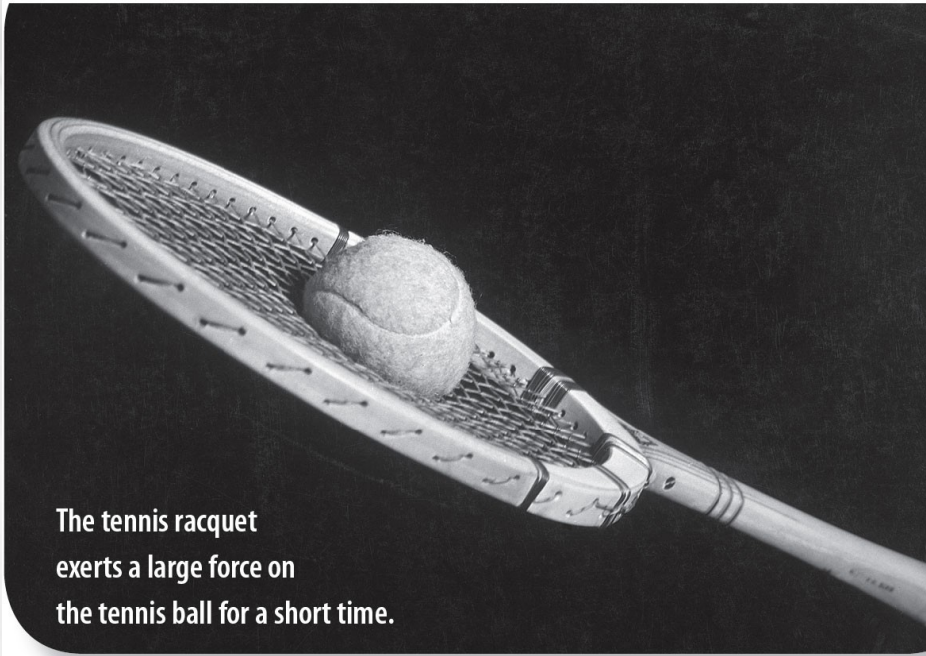


Why do you think you should have a good follow through?

You increase the time of contact for the same force F (same muscle). What happens to the speed of the ball ? (that is to its change of momentum ?)

Why a pitcher keep the baseball as long as possible when throwing ?

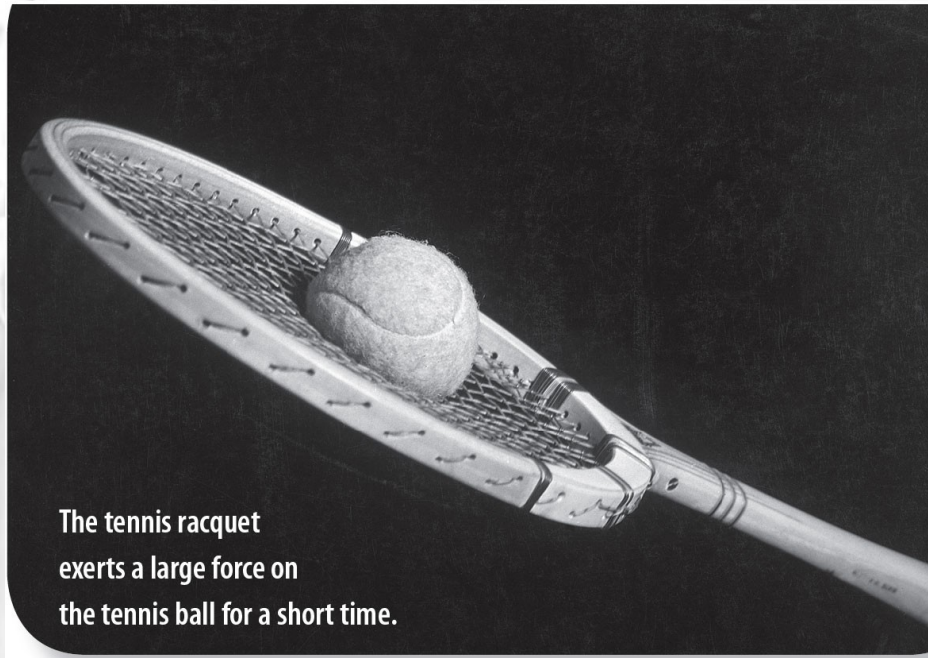
- This tells why we must exert a force to stop an object or get it to move.
 - To stop a moving object, we have to bring its momentum to zero.
 - To start moving an object, we have to impart some momentum to it.



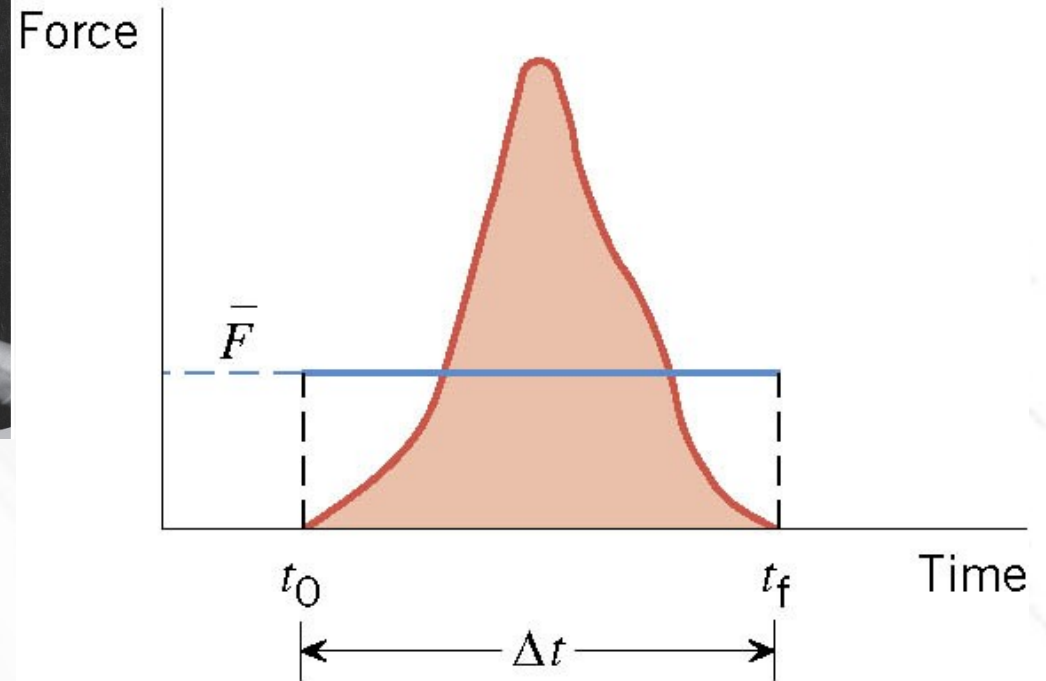
In reality the force applied to the Ball by the tennis is not constant . It increases and decreases. The formula is in fact:

$$m\Delta v = \overline{F} \Delta t$$

Change in momentum = impulse



$$m\Delta V = \overline{F}\Delta t$$



(b)

$\overline{F}$ Is the average force applied during the time Δt

IMPULSE-MOMENTUM THEOREM

And if more than 1 force act on an object, we need
To find first the net force before using the theorem
Called the impulse-momentum theorem;

When a net force acts on an object, the impulse of
this force is equal to the change in the momentum
of the object

impulse

$$\left(\sum \overline{F}\right)\Delta t = m(V_f - V_i) = mV_f - mV_i$$

final momentum

initial momentum

37

Momentum is mass x velocity. We will see that in more details next unit.

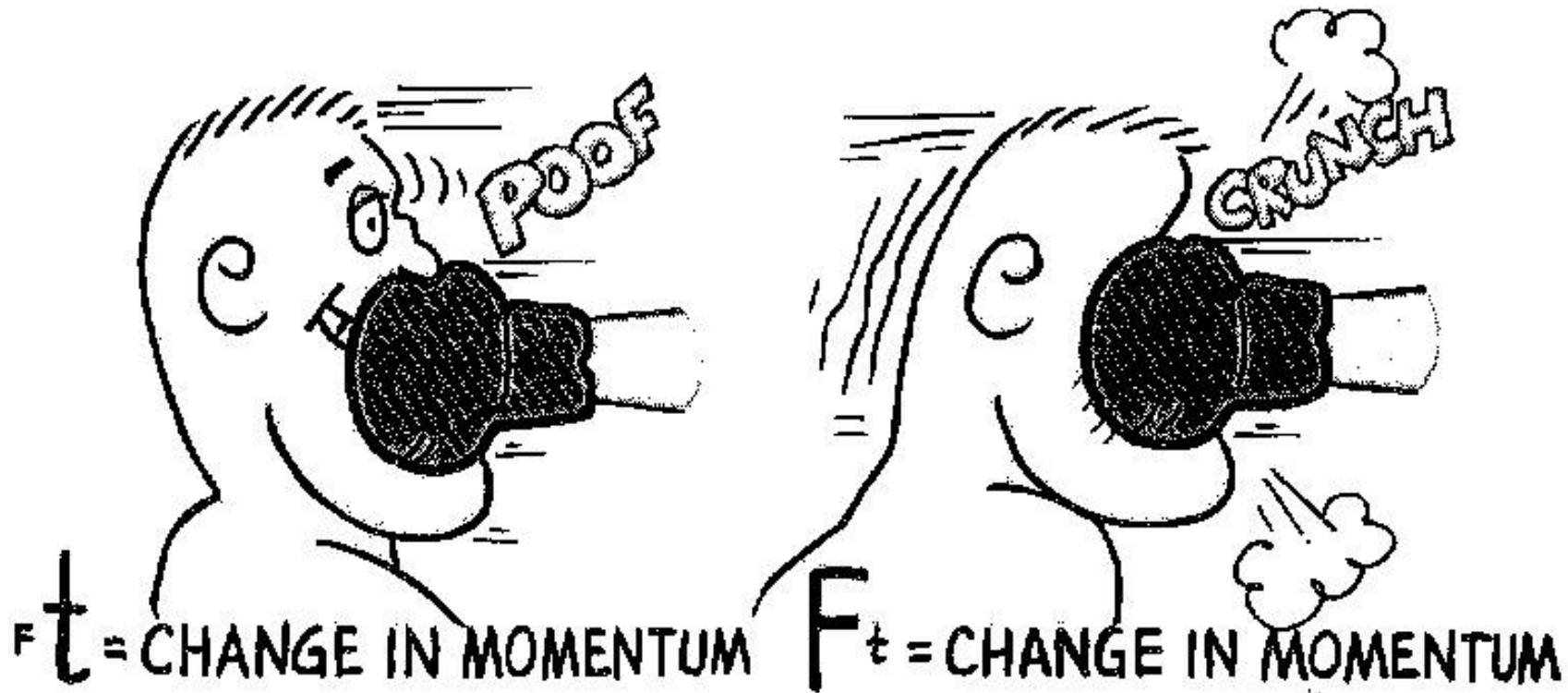
BUT LETS GO BACK TO THE BIG IDEA
behind this formula:

- It also tells us that we can change the motion using various forces and time intervals:

$$F\Delta t = m\Delta V$$

- Use a large force for a short time, or
- Use a small force for a long time.

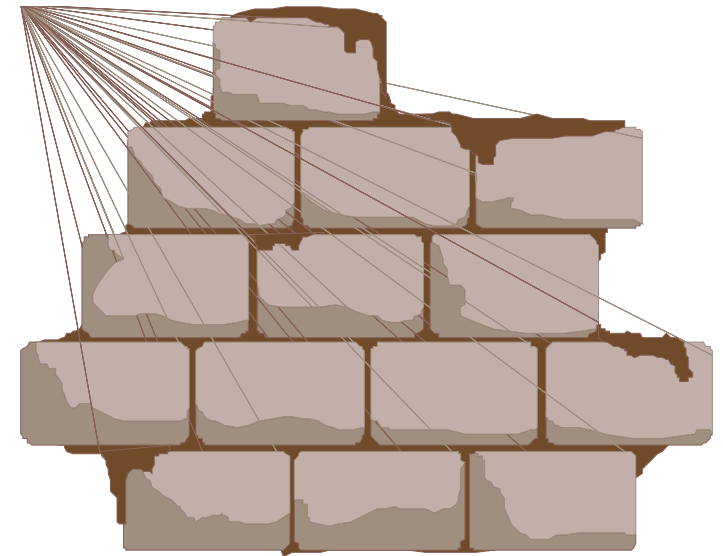
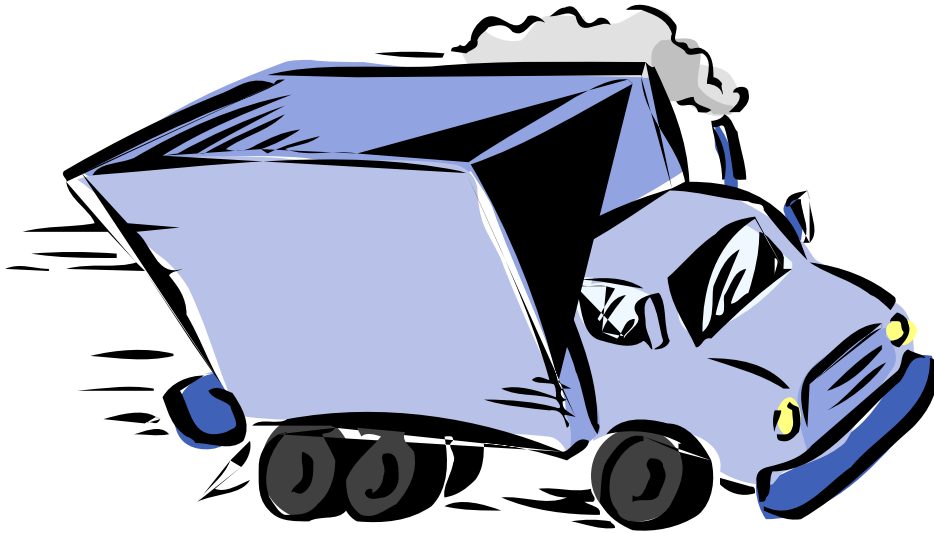
Change in motion = change in momentum



SO WHY DO YOU THINK a boxer that is getting punched needs to “roll over” ?

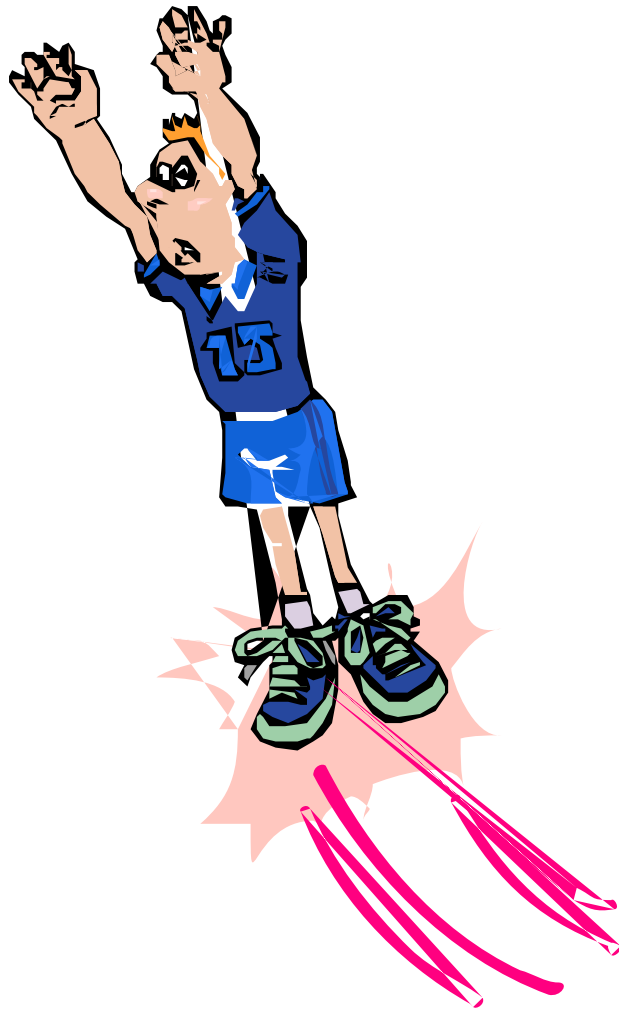
He moves away when the glove hits versus staying put ?

In both cases, the **change in momentum is the same**



A truck can be stopped by a haystack
Or by a brick of wall. **The change of motion
Is the same.** But what about time and force?
Which situation would you prefer ?

$$m\Delta v = \overline{F}\Delta t \quad \text{or} \quad m\Delta v = \overline{F}\Delta t$$



When you jump and you come
Back to the ground,
why don't you keep the
Legs stiff instead of bending them ?

Change in motion means a force is applied to the ground so
The ground applies the same force on the jumper.



www.shutterstock.com · 5256904

Large momentum (reach peak velocity)

→ large impulse

With large force and small time (hard part of hand)

- 1 To slow down a car, a braking force of 1 200 newtons is applied for 10 seconds. How much force would be needed to produce the same change in velocity in 6 seconds ?

- 2 A frictionless wagon going at 2.5 meters per second is pushed with a force of 380 N, and its speed increases to 6.2 meters per second in 4.0 seconds. What is its mass?

3. An average force of 300N acts for a time of 0.04s on a golf ball.
A) What is the magnitude of the impulse acting on the golf ball ?
B) What is the change in the golf's momentum ? (change in momentum)
C) What is the force the golf's ball acts on the club ?
4. A force of 35N acts on a ball for 0.2s . the ball is initially at rest:
A) What is the impulse on the ball ?
B) What is the change in momentum

Textbook 8.11 and fig 8.22

5. A force of 35N acts on a ball for 0.2s . the ball is initially at rest:

A)What is the impulse on the ball ?

B)What is the change in momentum

6.A 0.12kg ball traveling with a speed of 40m/s is brought to rest in a catcher's mitt.

What is the size of the impulse exerted by the mitt on the ball ?

What is the change in momentum ?

7. A 60kg front-seat passenger in a car moving initially with a speed of 18m/s (40mph) is brought

To rest by an air bad in a time of 4.0s .

A)What is the impulse acting on the passenger ?

B)What is the average force acting on the passenger in this process ?

8. What force is needed to accelerate a 60kg wagon from rest to 5m/s in 2 s?

9. more challenging: A spacecraft has 2 rocket engines, one producing a thrust of 300N and the other 750N. Firing the Smaller engine for 10s speeds the ship from 80m/s to 95m/s and the large engine for 12s, How fast is the ship then going ?

10. more challenging. A spacecraft is going 350m/s; a retro rocket that provides a force of 520N for 10s slows it down to 300m/s. Then another retro rocket fires , with a force of 130N for 4s. How fast is the craft then going.

11. A net force of 20N acting on a wooden block produced an acceleration of 4m/s/s for the block. What is the mass of the block? In class

12. A pulled tablecloth exerts a frictional force of 0.6N on a plate with a mass of 0.4kg. What is the acceleration of the plate ?

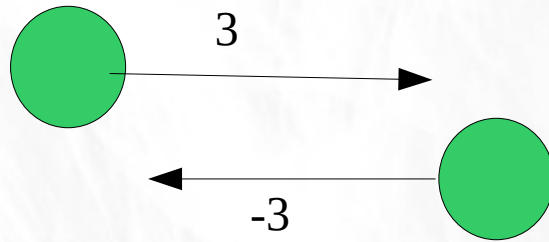
13. A single force of 40N acts upon a 5kg block. What is the magnitude of the acceleration of the block ?

14. If your weight is 600N

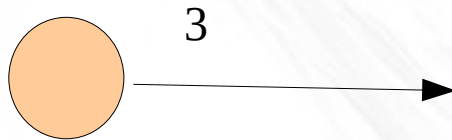
A) What is the mass in kg

B) what is the weight in lbs

Consider a green bouncing ball that bounces and an orange ball that sticks. The wall “ feels a force “



Suppose $m = 1\text{kg}$
 So it is easier to compute.
 Change = final - initial



0

Both undergo a change in momentum
 And they both have the same initial
 velocity.

Which one undergo the largest change
 in motion $m\Delta V = m(V_2 - V_1)$
 (change in momentum)

This one applied the largest force to the wall.
 (change in momentum = Impulse = force x time)

You can suppose $V_1 = 3$

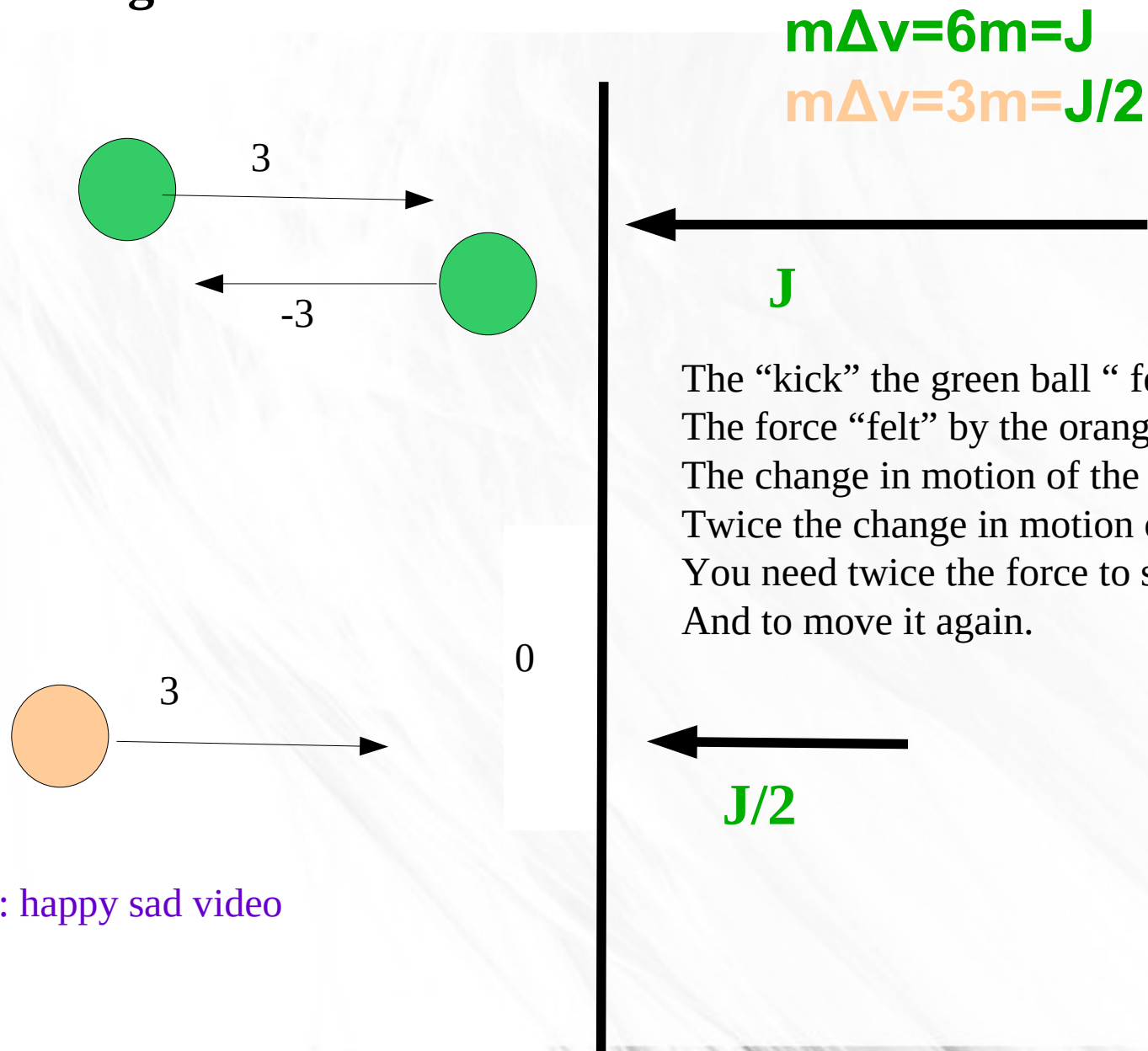
$V_2 = -3$ for green and $v_2 = 0$ for orange

$$m\Delta V = m(3 - (-3)) = \underline{\hspace{2cm}} \text{ (with } m = 1)$$

$$m\Delta V = m(3 - 0) = \underline{\hspace{2cm}} \text{ (with } m = 1)$$

**So which one “ feels” the greatest
 Impulse $J = F\Delta t$?**

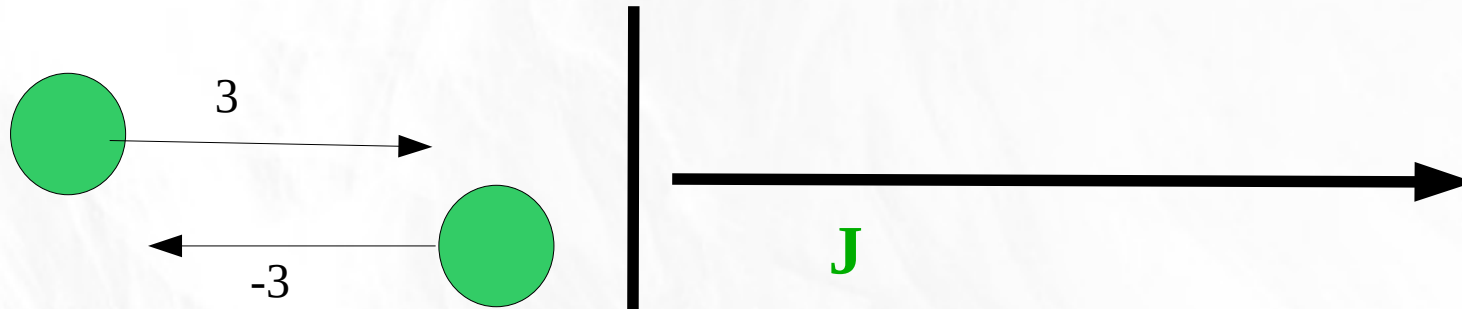
Consider a green bouncing ball that bounces and
an orange ball that sticks:



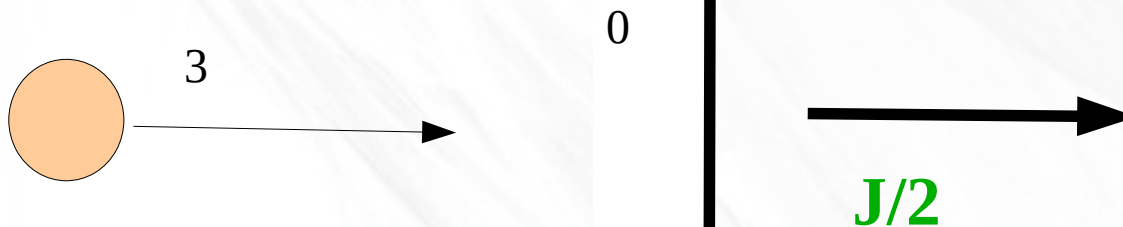
The “kick” the green ball “feels” is half
The force “felt” by the orange one.
The change in motion of the green is
Twice the change in motion of the orange.
You need twice the force to stop the ball
And to move it again.

Watch: happy sad video

Action-reaction = the wall “ feel” the same force F or $F/2$
Same magnitude but inverse direction.



Movie sad / happy ball

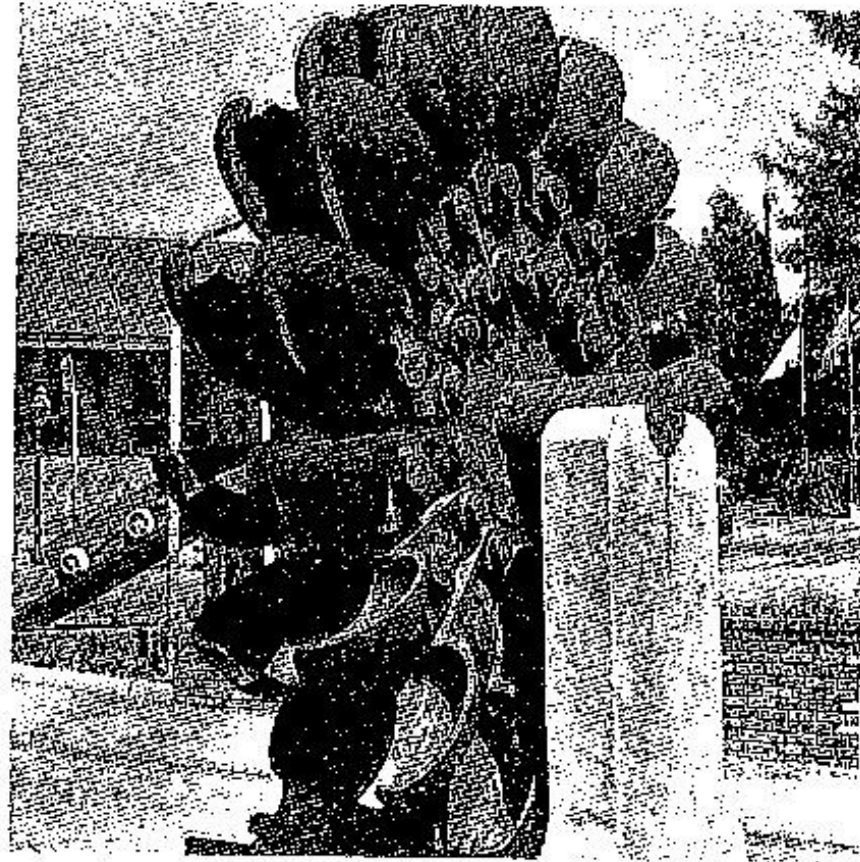
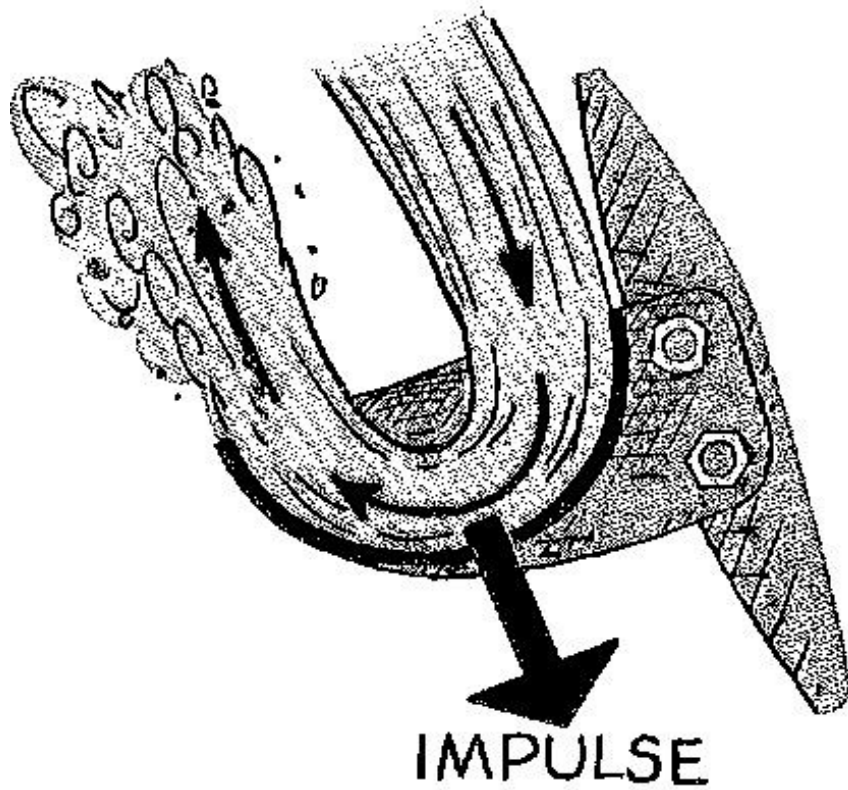


Newton's third law:

The ball acts upon the wall with the same force
The wall acts on them

So do you rather be hit by a ball that sticks to your forehead
Or by a bouncing ball ? (same mass and same initial velocity).

We suppose the ball is not a bullet that can go inside.

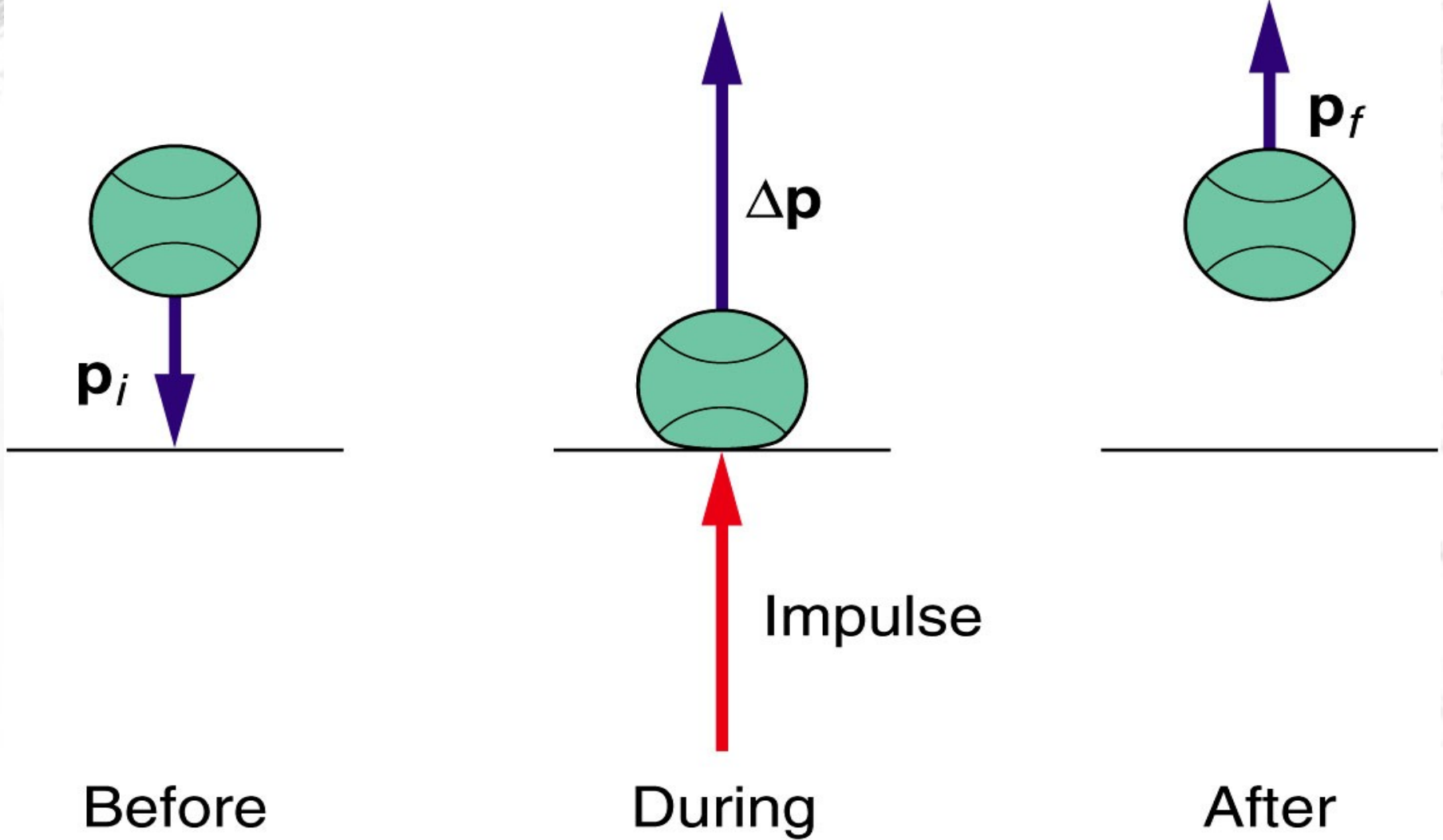


So why do you think this Pelton Wheel was a breakthrough for the Gold miners ?
 Before they were using flat paddles.
 (bouncing =
 more change in momentum = more impulse)
 Source: Paul Hewitt, conceptual Physics.

Pelton wheel

The Pelton wheel is an impulse-type water turbine. It was invented by Lester Allan Pelton in the 1870s. The Pelton wheel extracts energy from the impulse of moving water, as opposed to water's dead weight like the traditional overshot water wheel. [Wikipedia](#)





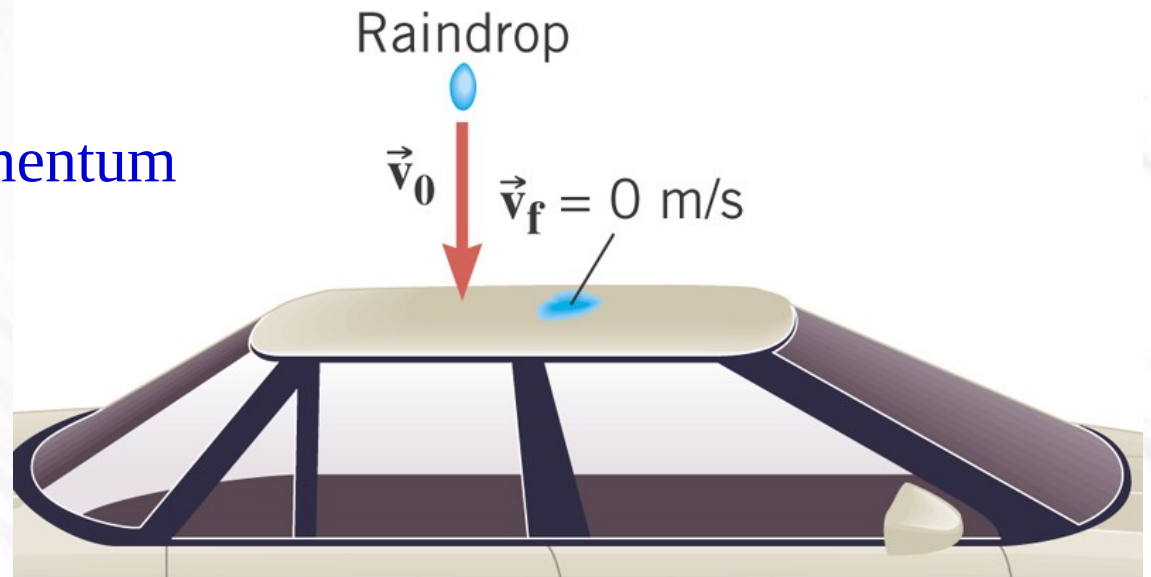
BOUNCING MEANS FORCE during time t (impulse) = change in momentum
It takes more impulse to catch and throw back a ball than to just stop it.

- **How does impulse and force differ ?**
- **What are the two ways impulse exerted on something can be increased ?**
- **For the same force, which cannon imparts the greater speed to a cannonball:
A long cannon or a short one ? Explain**
- **Why is it a good idea to have your hand extended forward when you are getting
Ready to catch a fast-moving baseball in your bare hand?**
- **Which case illustrates the greatest change in momentum :**
 - (A) a baseball being caught**
 - (B) a baseball being being thrown**
 - (C) or being caught and thrown back ?**
- **in the preceding question, in which case is the greatest impulse required?**
- **why impulses are greater when bouncing takes place**

From your book :

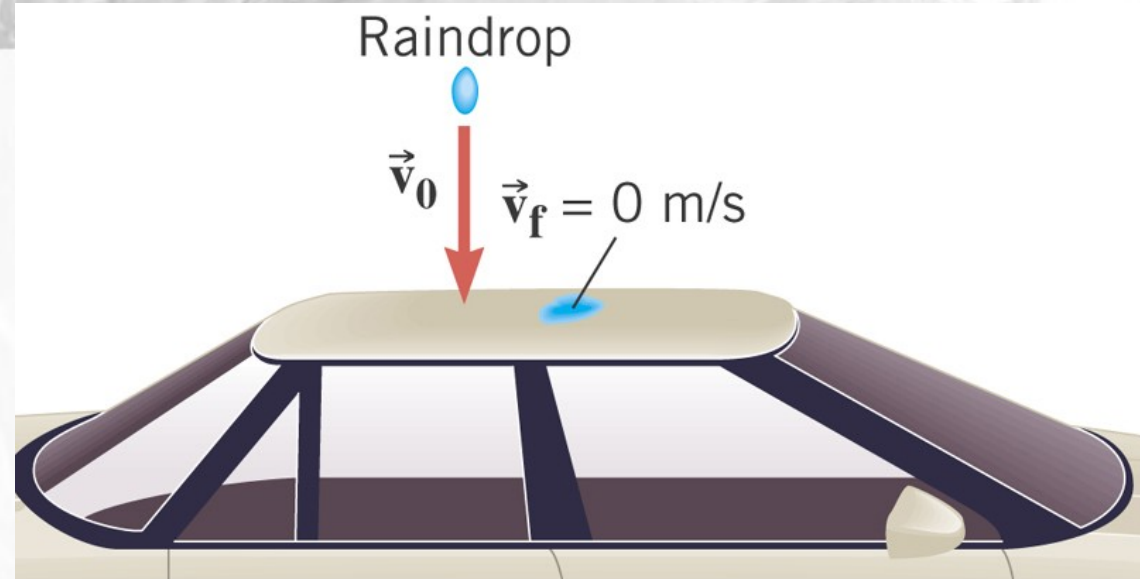
Rain comes down with a velocity of -15 m/s and hits the roof of a car. The mass of rain per second that strikes the roof of the car is 0.060 kg/s . Assuming that rain comes to rest upon striking the car, find the average force exerted by the rain on the roof.

Force x time = change in momentum
 $F t = m (V_2 - V_1)$



7.1 The Impulse-Momentum Theorem

Neglecting the weight of the raindrops, the net force on a raindrop is simply the force on the raindrop due to the roof.



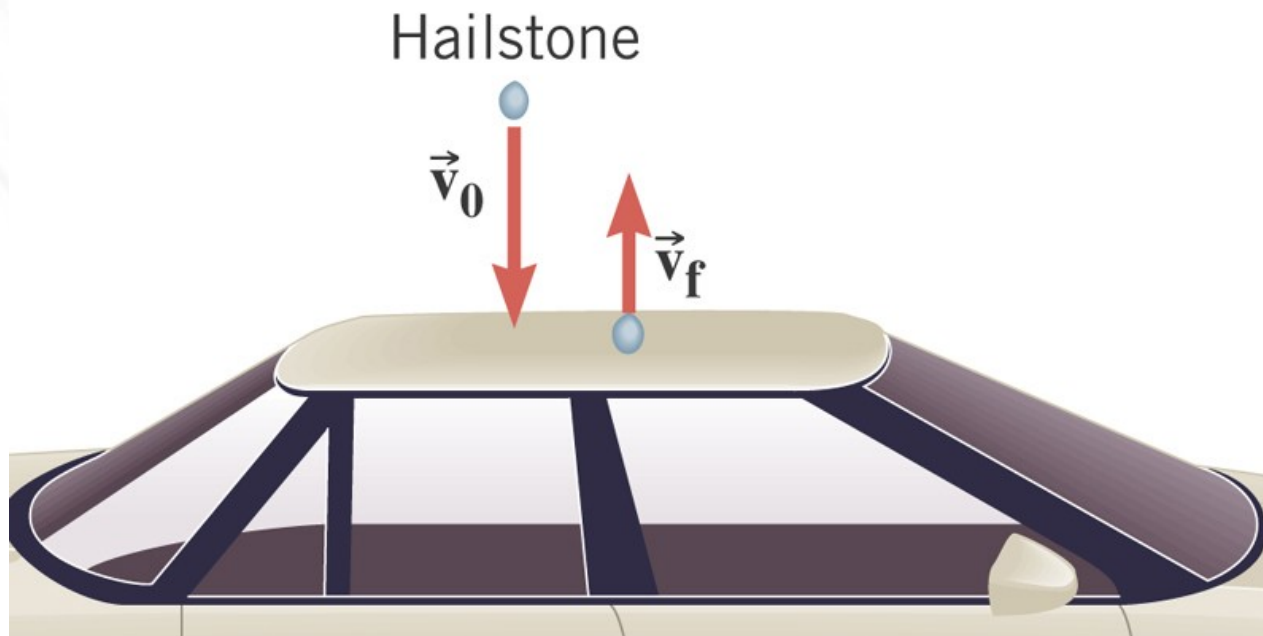
$$\vec{\mathbf{F}} \Delta t = m \vec{\mathbf{v}}_f - m \vec{\mathbf{v}}_o \quad \longrightarrow \quad \vec{\mathbf{F}} = -\left(\frac{m}{\Delta t}\right) \vec{\mathbf{v}}_o$$

$$\vec{\mathbf{F}} = -(0.060 \text{ kg/s})(-15 \text{ m/s}) = +0.90 \text{ N}$$

Conceptual Example 3 Hailstones Versus Raindrops

Instead of rain, suppose hail is falling. Unlike rain, hail usually bounces off the roof of the car.

If hail fell instead of rain, would the force be smaller than, equal to, or greater than that calculated in Example 2?



1. A ball traveling with an initial momentum of 2.5 km kg/s bounces off and comes back in the opposite

Direction with a momentum of -2.5 kg m/s

A) What is the change in momentum of the ball ?

B) What impulse would be required to produce this change ?

2. A ball traveling with an initial momentum of 4.0 kgm/s bounces back in the opposite direction with a

Momentum of -3.5 kgm/s

A) What is the change in momentum of the ball ?

B) What impulse is required to produce this change ?

Newton's Second : applications

$$F\Delta t = m\Delta V \quad F = \frac{m\Delta V}{\Delta t} \quad F = ma$$

When a net external force acts on an object of mass m , the acceleration that results is directly proportional to the net force and has a magnitude that is inversely proportional to the mass. The direction of the acceleration is the same as the direction of the net force.

$$\vec{\mathbf{a}} = \frac{\sum \vec{\mathbf{F}}}{m} \quad \sum \vec{\mathbf{F}} = m\vec{\mathbf{a}}$$

SI Unit for Force

$$(\text{kg})\left(\frac{\text{m}}{\text{s}^2}\right) = \frac{\text{kg} \cdot \text{m}}{\text{s}^2}$$

This combination of units is called a *newton* (N).

4.3 *Newton's Second Law of Motion*

Table 4.1 **Units for Mass, Acceleration, and Force**

System	Mass	Acceleration	Force
SI	kilogram (kg)	meter/second ² (m/s ²)	newton (N)
CGS	gram (g)	centimeter/second ² (cm/s ²)	dyne (dyn)
BE	slug (sl)	foot/second ² (ft/s ²)	pound (lb)

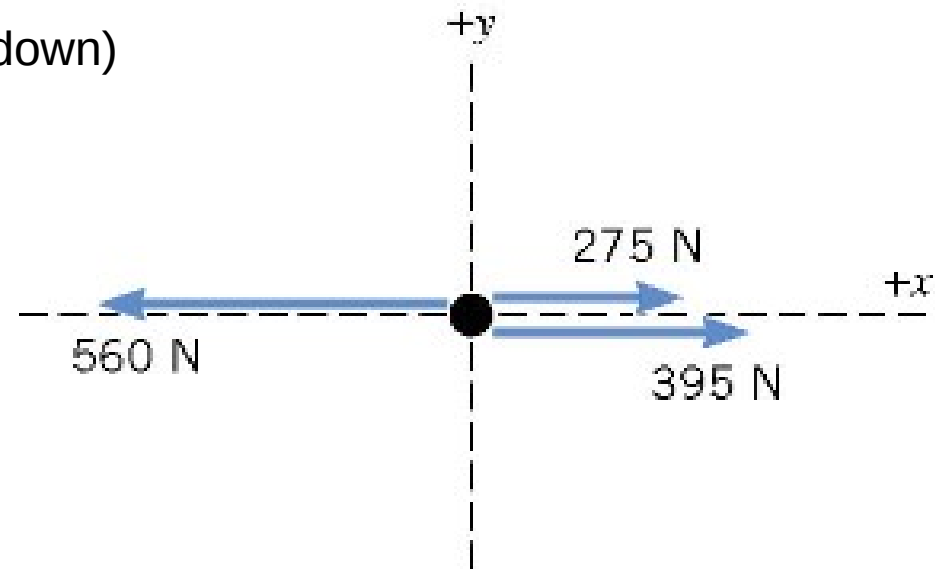
A **free-body-diagram** is a diagram that represents the object and the forces that act on it.

Remember $F_{\text{net}} = \text{mass} \times \text{acceleration}$

F_{net} is sum of forces (right – left or up – down)



(a)



(b) Free-body diagram of the car

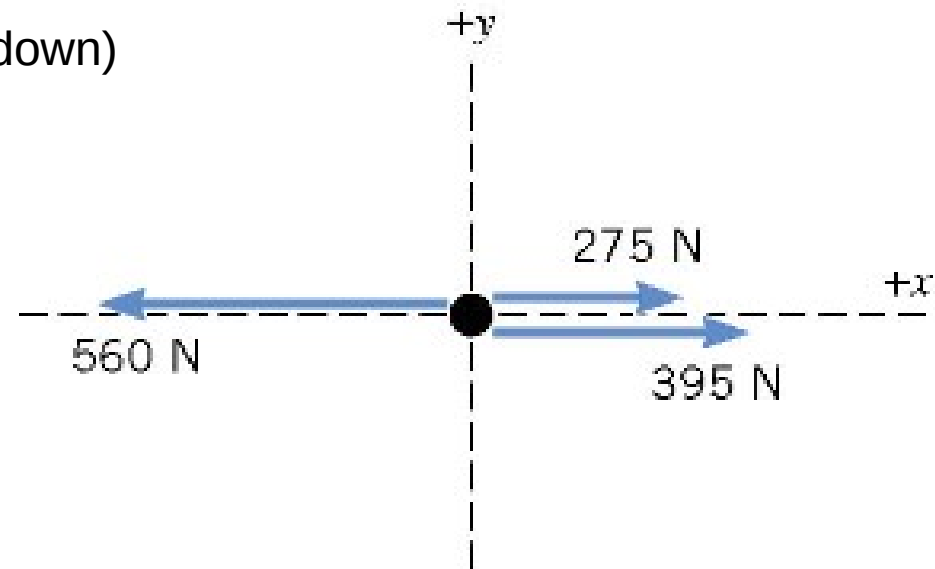
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(a)

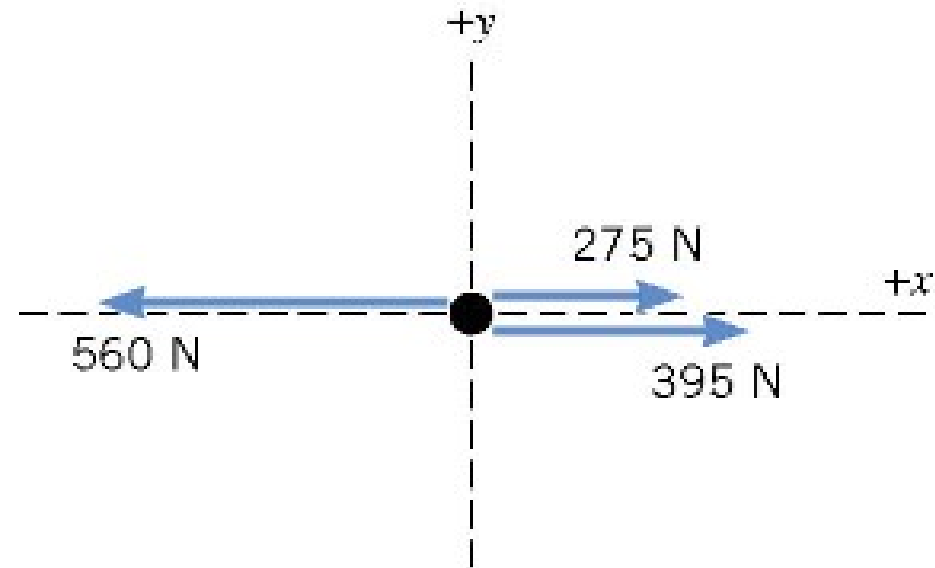


(b) Free-body diagram of the car

Newton's Second Law of Motion



(a)



(b) Free-body diagram of the car

The net force in this case is:

$$275 \text{ N} + 395 \text{ N} - 560 \text{ N} = +110 \text{ N}$$

and is directed along the + x axis of the coordinate system.

If the mass of the car is 1850 kg then, by Newton's second law, the acceleration is

$$a = \frac{\sum F}{m} = \frac{+110 \text{ N}}{1850 \text{ kg}} = +0.059 \text{ m/s}^2$$

The direction of force and acceleration vectors can be taken into account by using x and y components.

$$\sum \vec{F} = m\vec{a} \quad F_{net} = m\vec{a}$$

is equivalent to:

$$\sum F_y = ma_y$$

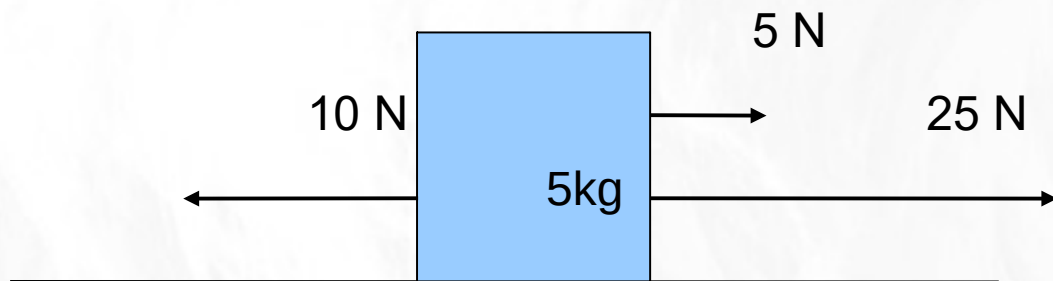
$$\sum F_x = ma_x$$

$$F_{net_y} = ma_y$$

$$F_{net_x} = ma_x$$

Or up - down = ma_y
(up is positive)

Or right - left = ma_x
(right is positive)



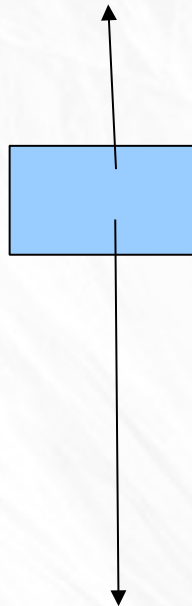
Find the acceleration

Find the net force

Or **right - left = ma_x**
(right is positive)

At a given instant time, a 4kg rock that has been dropped from a high cliff experiences a force of air resistance of 15N.

What are the magnitude and direction of the acceleration of the rock/
(Do not forget the gravitational force = weight of the rock)



It accelerates @ down. So take down to be Positive.

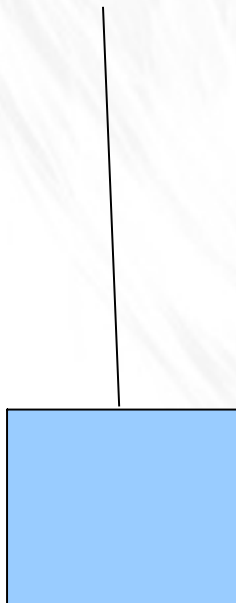
Down – up = mass x acceleration

Net force is down → acceleration is down

An upward force of 18N is applied via a string to lift a ball with a mass of 1.5 kg.

A)What is the net force acting upon the ball?

B)What is the acceleration of the ball ?



Draw a free body diagram

Always take the positive direction
= same direction as acceleration

Or up - down = ma

(up is positive)

Remember $F_{\text{net}} = \text{mass} \times \text{acceleration}$

F_{net} is sum of forces (right – left or up – down)

A rope exerts a constant force of 250N to pull a 60kg crate across the floor. The velocity of the crate is observed to increase from 1m/s to 3m/s in a time of 2 seconds under the influence of this force and the frictional force exerted by the floor on the crate. Suppose it is moving @ right. Always acceleration >0

A) What is the acceleration of the crate

(hint: kinematics. What is the definition of acceleration?. Change of speed over time)

B) What is the total force acting upon the crate ?

(use Newton's second's law $F_{\text{net}} = \text{mass} \times \text{acceleration}$)

C) What is the magnitude of the frictional force acting on the crate?

(right – left = mass $\times$ acceleration)

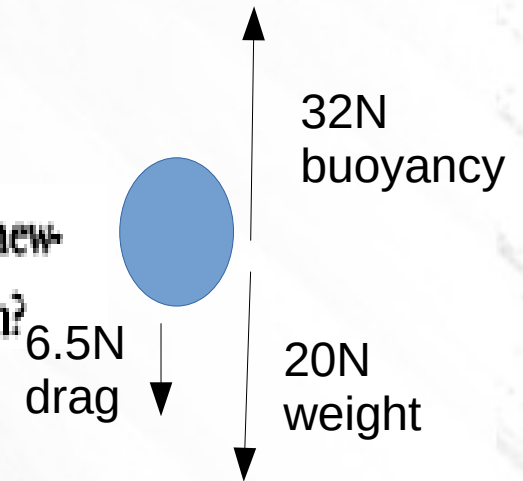
D) What force would have to be applied to the crate by the rope for the crate to move with a constant velocity / explain./

Take up to be positive and right to be positive

1 What thrust is needed to fire a 350-kilogram rocket straight up with an acceleration of 8.0 meters per second squared?

2 A 2.0-kilogram weather balloon is released and begins to rise against 6.5 newtons of viscous drag. If its buoyancy is 32 newtons, what is its acceleration?

Up – down = mass x acceleration



3 What is the acceleration of a 1 200-kilogram boat if its motor produces 8 500 newtons of forward thrust and the viscous drag is 6 200 newtons?

right – left = mass x acceleration

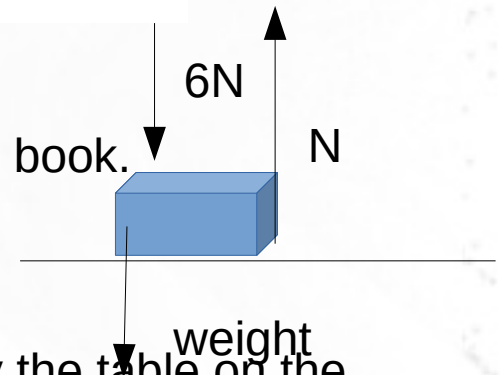
4

What braking force is needed to bring a 2 200-kilogram car going 18 meters per second to rest in 6.0 seconds?

5) A 0.4kg book rests on a table. A downward force of 6 N is exerted on the top Of the book by a hand pushing down on the book.

A) What is the magnitude of the gravitational force acting upon the book ?

B) What is the magnitude of the upward (normal) force exerted by the table on the book ? If the book is in static equilibrium (no acceleration so up = down or $up - down = 0$)



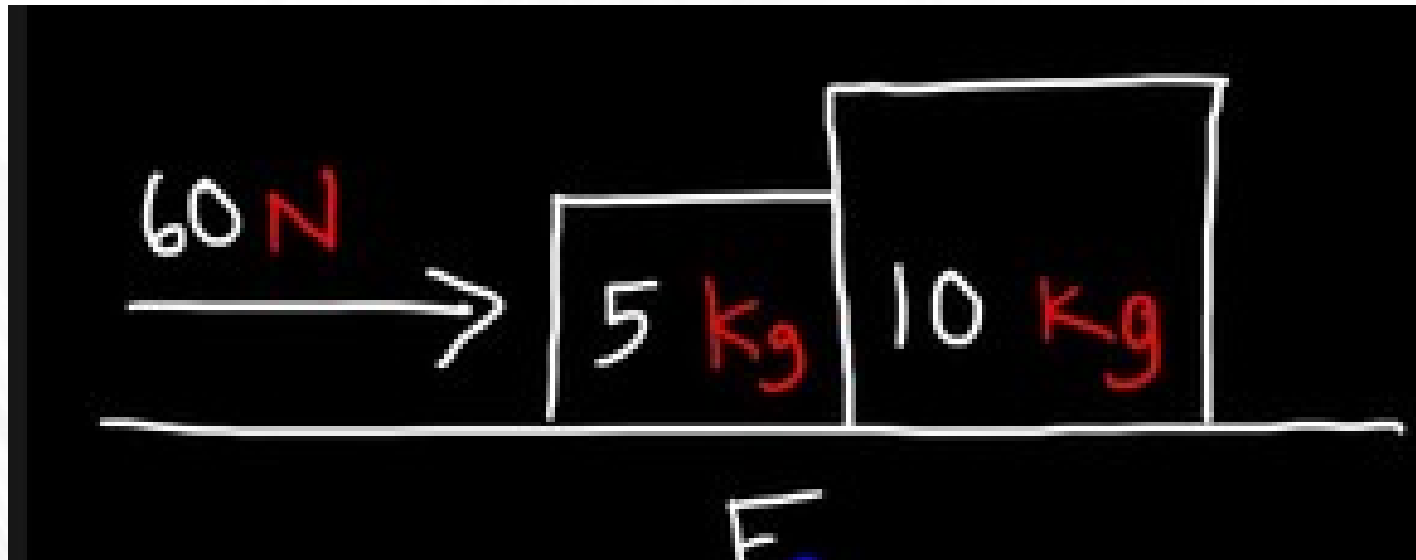
6) An upward force of 18N is applied via a string to lift a ball with a mass of 1.5kg. What is the acceleration of the ball ? (up – down = mass x acceleration)

7) A constant horizontal force of 30N is exerted by a string attached to a 5 kg block being pulled Across A table. The block also experienced a frictional force of 5N due to contact with the table.

A) What is the horizontal acceleration of the block ? (right – left = mass x acceleration)

B) If the block starts from rest , what will be its velocity be after 3s ? (use kinematics equations)

C)How far will it travel in these 3 seconds ? (kinematics equation)



Find the acceleration a and the contact forces between the block.

Step1 → Have the 2 masses in a “bag” / So now the system has a mass of 15kg. What is the acceleration?

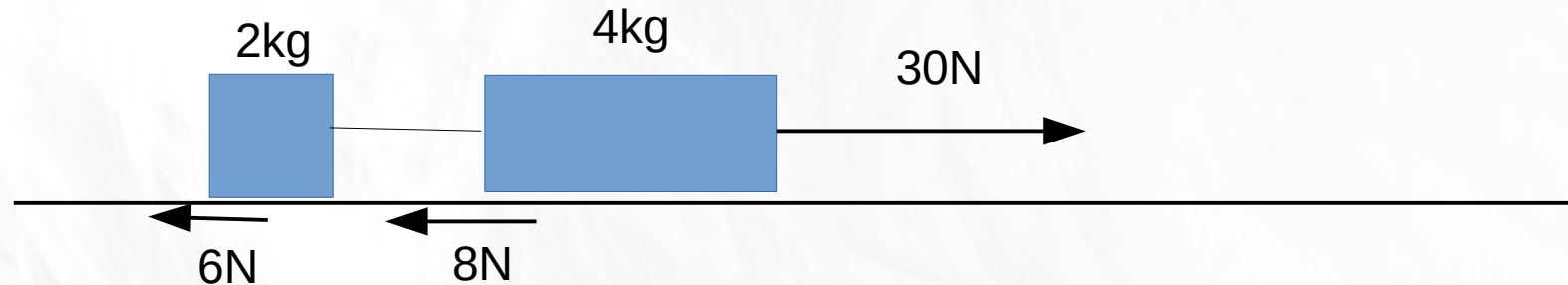
Step2: have just the 10kg in a “bag”. That’s your new system. What is the Force acting on it? What is its acceleration (step1) ? Find the contact force $F_{5 \rightarrow 10}$

Step 3: verify. Now the system is just the 5kg. What are the forces acting on it? Remember Newton’s third law. $F_{5 \rightarrow 10} = - F_{10 \rightarrow 5}$. Find the acceleration. Should be the same

70

<https://www.youtube.com/watch?v=lin29KFE4So>

8)



2 blocks tied together by a horizontal string are being pulled across the table by a horizontal force of 30N as shown. The 2 kg block has a 6N frictional force exerted on it by the table, and the 4kg block has an 8N frictional force acting on it.

A) What is the net force acting on the entire two-block system ?

(take the whole 6kg system and compute right – left = F_{net})

B) What is the acceleration of the system ? ($F_{\text{net}} = ma$ or right – left = ma)

C) What force is exerted on the 2kg block by the connecting string ? (Cut out this 2kg block and consider only the forces Acting on this block. Its acceleration is the same as that of the entire system).

D) Find the net force acting on the 4 kg and calculate its acceleration. How does this value Compare to that found in B) ?

9) A dish with a mass 0.4kg has a force of kinetic friction of 0.15N exerted on it by a moving Tablecloth for a time of 0.2s .

A) What is the acceleration of the dish ? ($F = m a$)

B) What velocity does it reach in this time , starting from rest ?

C) How far (in cm) does the dish move in this time?

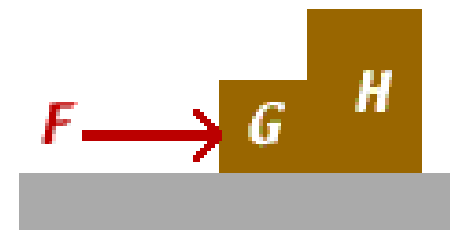
12) Two cardboard boxes full of books are in contact with each other on a frictionless table. Box H has twice the mass of box G. G has a mass of 5 kg . You push on box G with a horizontal force $F = 10\text{N}$. I will guide you so you can find the Force experienced by H.

1) Find the acceleration of the whole system

2) Now just consider the box G/ Its acceleration is the same as found In 1). The forces acting on it are : the push and the recoil force from H F' .

(G pushes on H so H pushes back on it). Solve for F' .

3) Now you can find the force experienced by H (same as F')

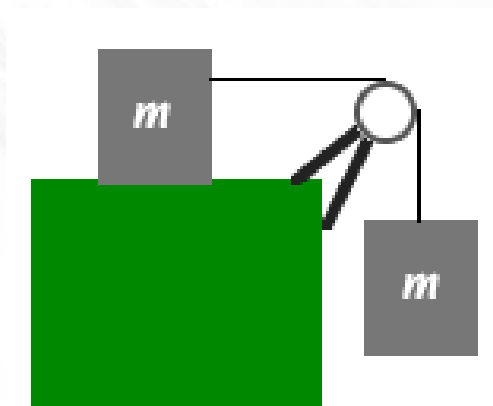


11) This system of 2 blocks is moving and speeding up @ down. There is No fiction involved and the goal is to find the acceleration of the system. I will guide you. The acceleration due to gravity is g .

1) Only consider the block sliding on the table at an acceleration noted a . The only force acting on it is T the tension of the string due to the other block. Find the acceleration a of the block as a function of m and T .

2) Now consider the other block hanging. It has the same acceleration but it is Experiencing 2 forces. Its weight mg and T (action reaction). Use your result from 1) to find a as a function of g .

3) conclusion ? Acceleration is larger ? Smaller ? The same as $g = 9.8$?



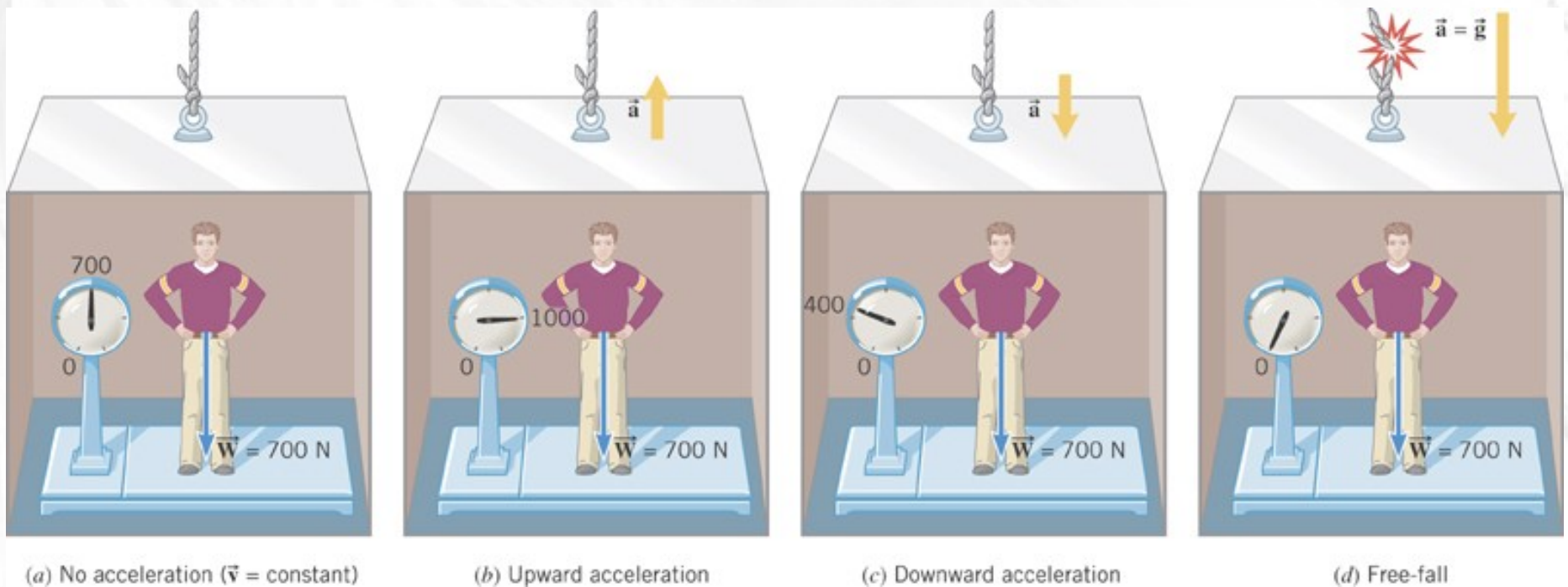
Gravity and weightlessness



Apparent Weight

The apparent weight of an object is the reading of the scale.

It is equal to the normal force the man exerts on the scale.



1 → equilibrium

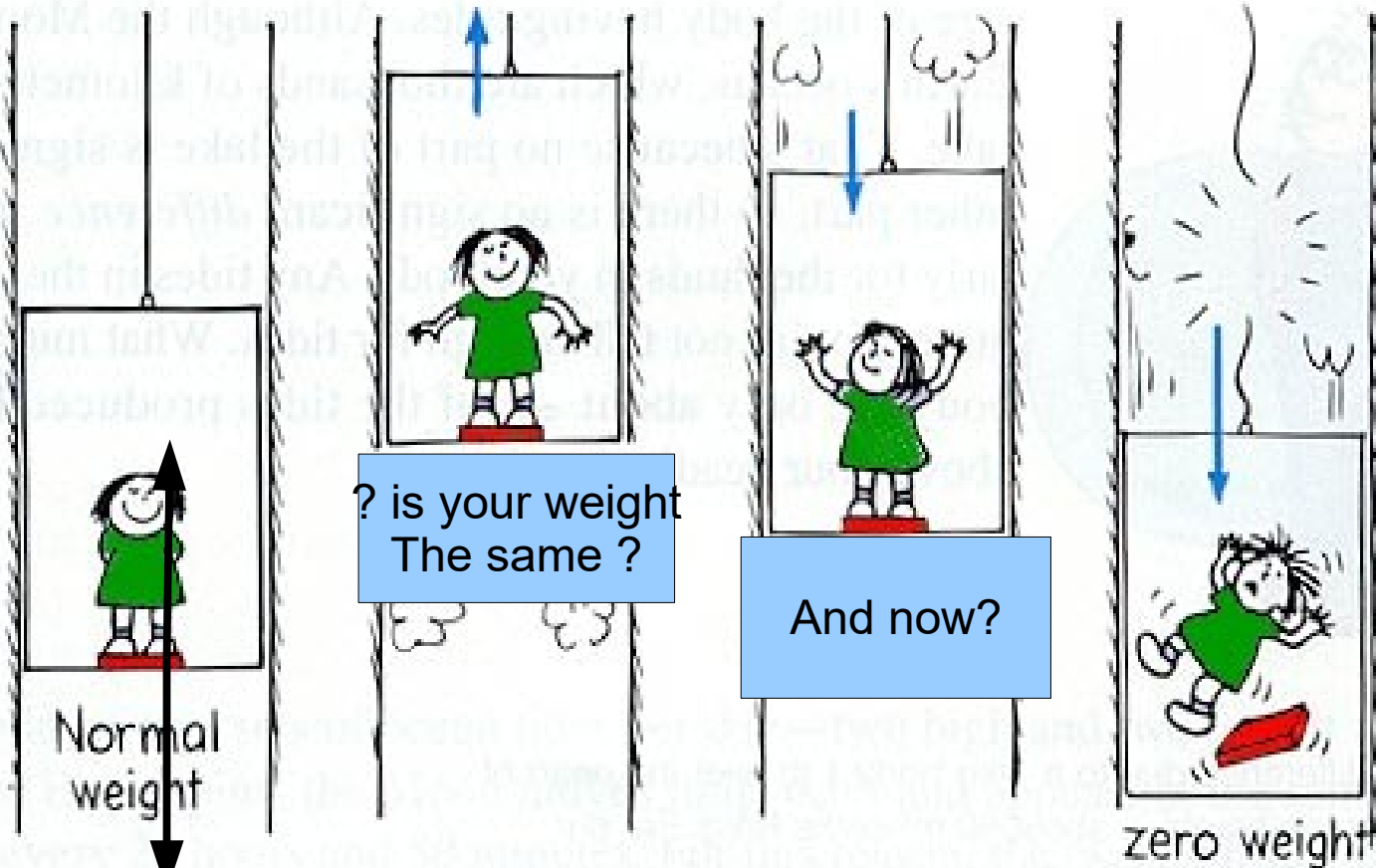
2 → up > down a is @ up because F_{net} @ up

Apparent weight is the normal force = reading from bathroom scale > weight

3 → down > up a is @ down because F_{net} @ down . Normal < weight

4 → $N = 0$ → apparent weight is 0. There is no support. 0 on bathroom scale

She feels weightless (but your weight is not 0)



The weight pulls the girl and
The ground pushes the girl up
This is the normal force. You feel
The normal force not the weight..

The weight pulls the girl but
There is no normal force.
So the girl does not "feel" her weight

A 60kg woman in an elevator is accelerating upward at a rate of 1.2m/s^2 .
OUR system is just the woman. We are just considering the woman and the Forces acting on her.

A) Just using Newton's second's law.

What is the net force acting upon the woman ? (use $F_{\text{net}_y} = ma_y$)

B) Draw all the forces acting on the woman.

C) What is the gravitational force acting upon the woman / (weight = mg)

D) What is the normal force pushing upward on the woman feet

Hint: up – down = ma with up is the support force (normal force) and down is the weight,

E) If the woman were standing on a scale, the scale would read this normal force.

A scale reads whatever force you are exerting on it.

This is called her apparent weight. What is her apparent weight?



Textbook 4.9

<https://www.youtube.com/watch?v=sVVKpRvuNG0>

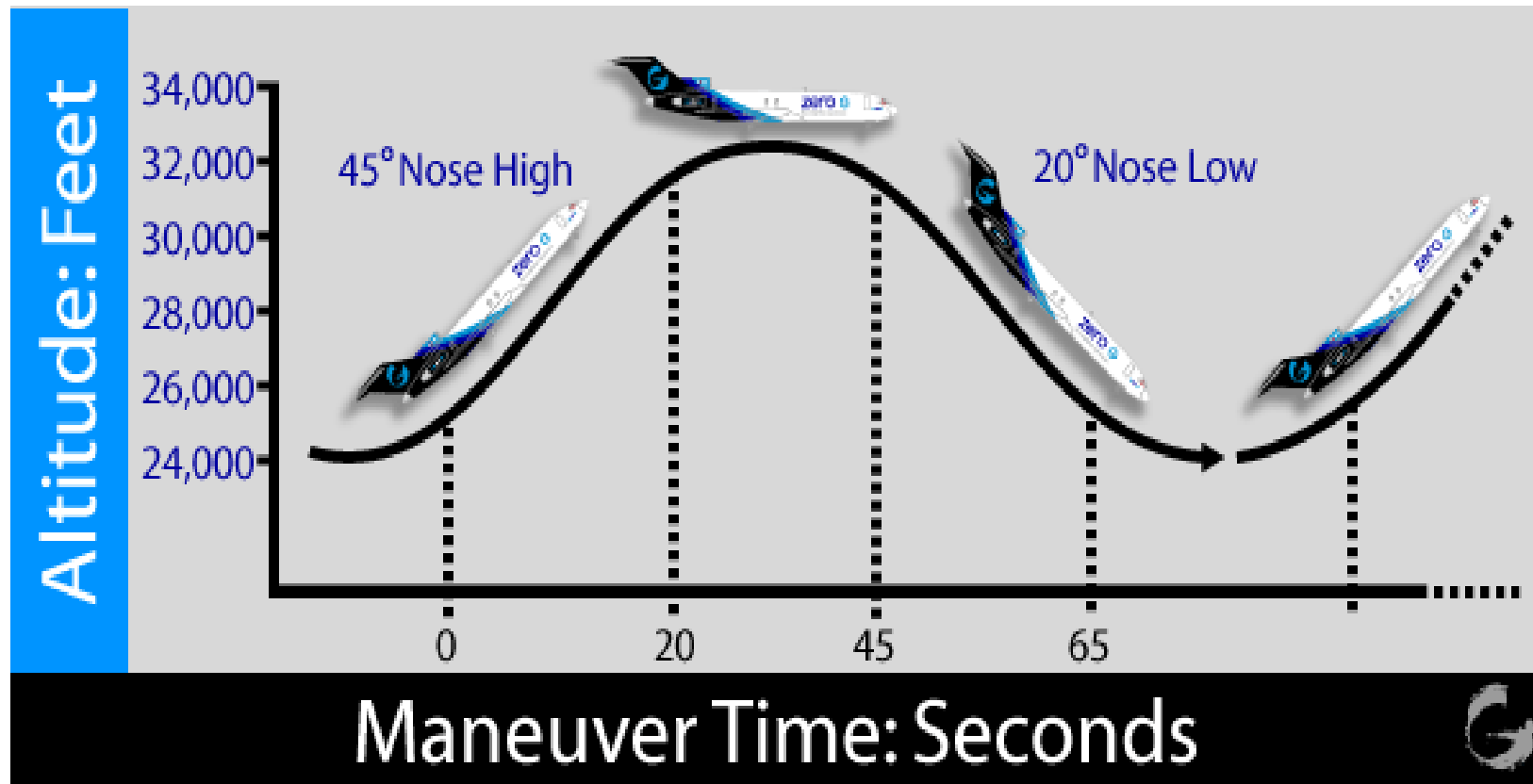
Spinning in zero gravity

http://www.youtube.com/watch?v=7tEkKhMVoS4&feature=player_embedded

http://www.youtube.com/watch?v=H0pGEq7bhLM&feature=player_embedded

<https://www.youtube.com/watch?v=LWGJA9i18Co>

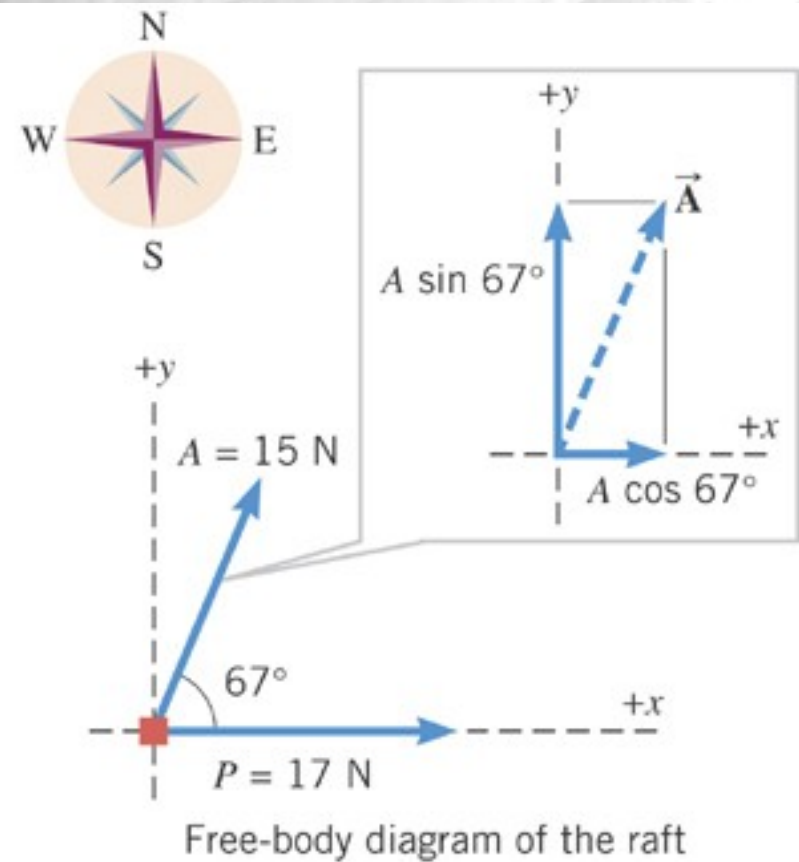
Zero gravity



While going down the plane and its inhabitants are in zero gravity for 10-15 seconds.



Mass is 1300kg



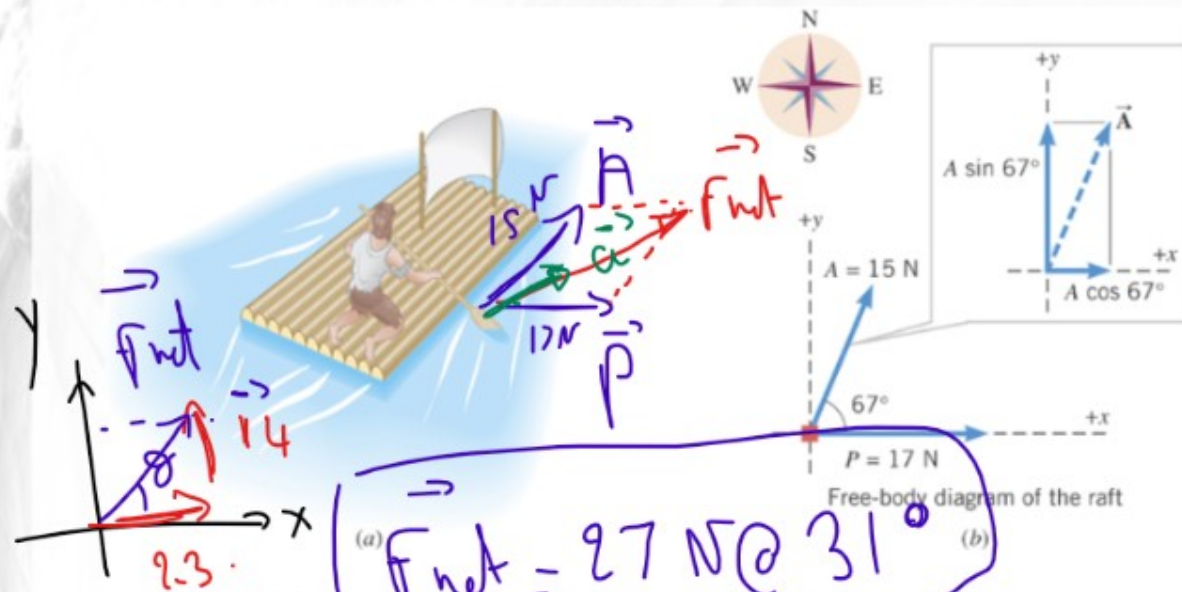
Free-body diagram of the raft

(b)

What is the net force? F_{net} ?

The acceleration is in the same direction as F_{net}

- 1) write the forces in standard notation
- 2) Use a table to find the the components of each force
- 3) Add the x-components to find the x-component of the net force F_x
- 4) Add the y-components to find the y-component of the net force F_y
- 5) Trace the net force F (use its components)
- 6) Find the magnitude and the direction of F
- 7) $F = \underline{\hspace{1cm}}\text{ N @ } \underline{\hspace{1cm}}$
- 8) since $a = F / \text{mass}$ divide the magnitude of F to find the magnitude of a
- 9) $a = \underline{\hspace{1cm}}\text{ m/s/s @ } \underline{\hspace{1cm}}$ (same direction)
- 10) you can find the components of a . Those are the acceleration along x or y



$$\vec{A} = 15\text{ N} @ 67^\circ$$

$$\vec{P} = 17\text{ N} @ 0^\circ$$

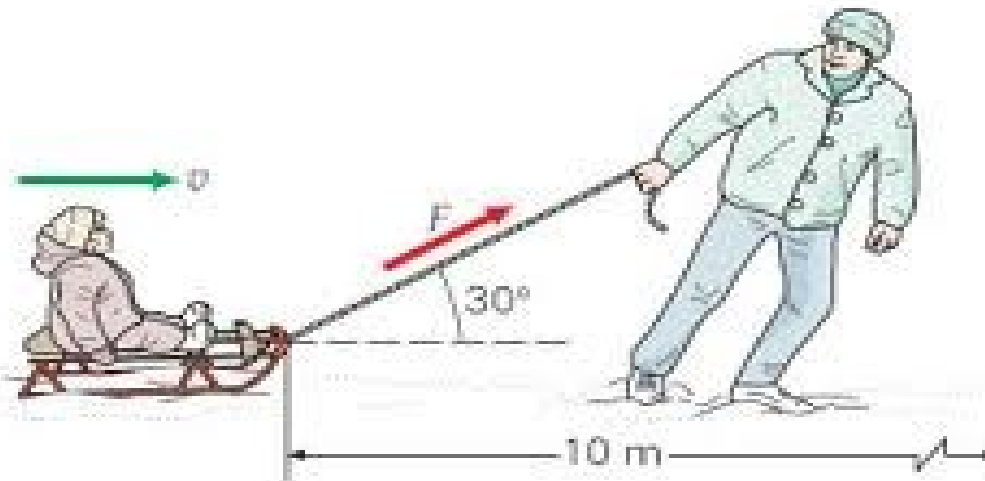
	x	y
A	6 N	14 N
P	17	0
$\vec{F}_{\text{net}}$	<u>23</u>	<u>14</u>
	F_x	F_y

$$\tan \theta = \frac{14}{23}$$

Find the acceleration along x
 The acceleration along y
 The total acceleration along its direction of motion
 : Mass is 1300kg

$$\vec{a} = \frac{\vec{F}_{\text{net}}}{m_{\text{raft}}} = (.02 \text{ m/s}^2 @ 31^\circ)$$

What is the acceleration of the child on the sled, with combined mass 40 kilograms, if the friction is 60 newtons and the rope is being pulled with a force of 170 newtons at an angle of 35° with the ground?



The sled is pulled over the distance 10m.
It starts from rest. What is the final speed of the sled ?

10) A sky diver has a weight of 750N. Suppose the air resistance force acting on the diver increases in Direct proportion to his velocity such that for every 10m/s that the driver's velocity increases, the force Of air resistance increases by 100N.

A) What is the net force acting on the sky diver when his velocity is 40m/s

B) What is the acceleration of the diver at this velocity ?

C) What is the terminal velocity of the sky diver (no acceleration, constant speed)

D) What would happen to the velocity of the sky diver of for some reasons (a draft) his velocity Exceeded the terminal speed ? Explain ?

Centripetal Force

- Acceleration:

- Magnitude:

$$a_c = \frac{v^2}{r}$$

- Direction: toward the center of the circle of motion

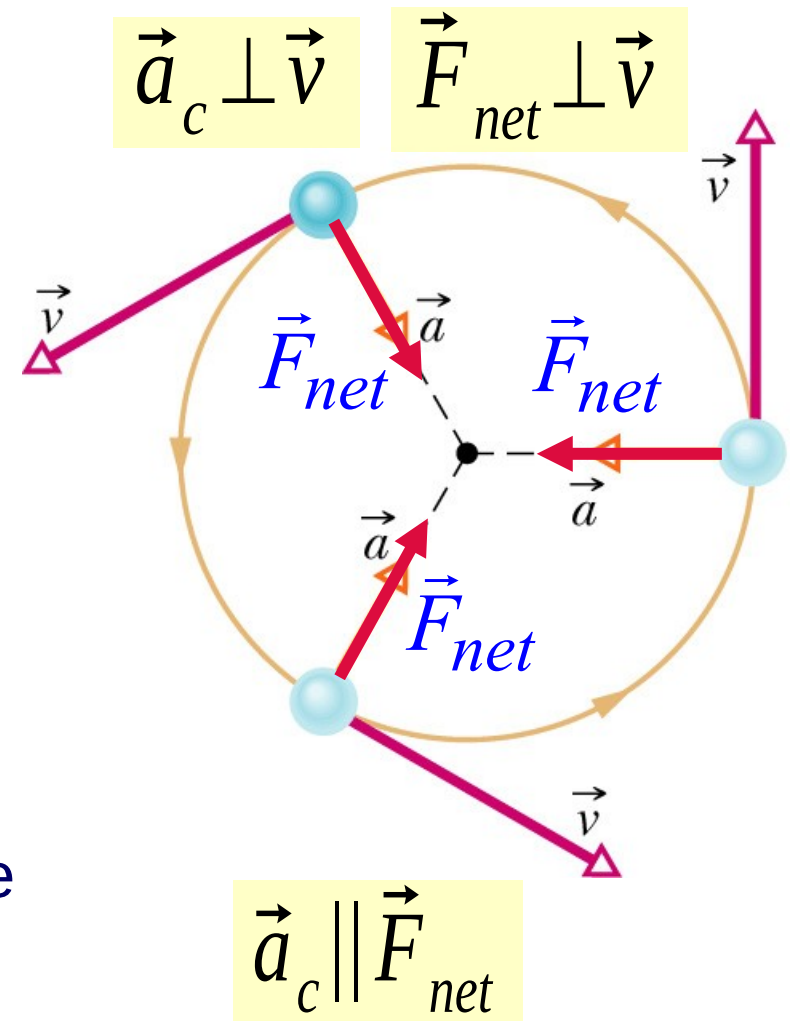
- Force:

- Start from Newton's 2nd Law

- Magnitude: $\vec{F}_{net} = m \vec{a}$

$$F_{net} = m a_c = \frac{mv^2}{r}$$

- Direction: toward the center of the circle of motion



5.3 Centripetal Force

Thus, in uniform circular motion there must be a net force to produce the centripetal acceleration.

The centripetal force is the name given to the net force required to keep an object moving on a circular path.

The direction of the centripetal force always points toward the center of the circle and continually changes direction as the object moves.

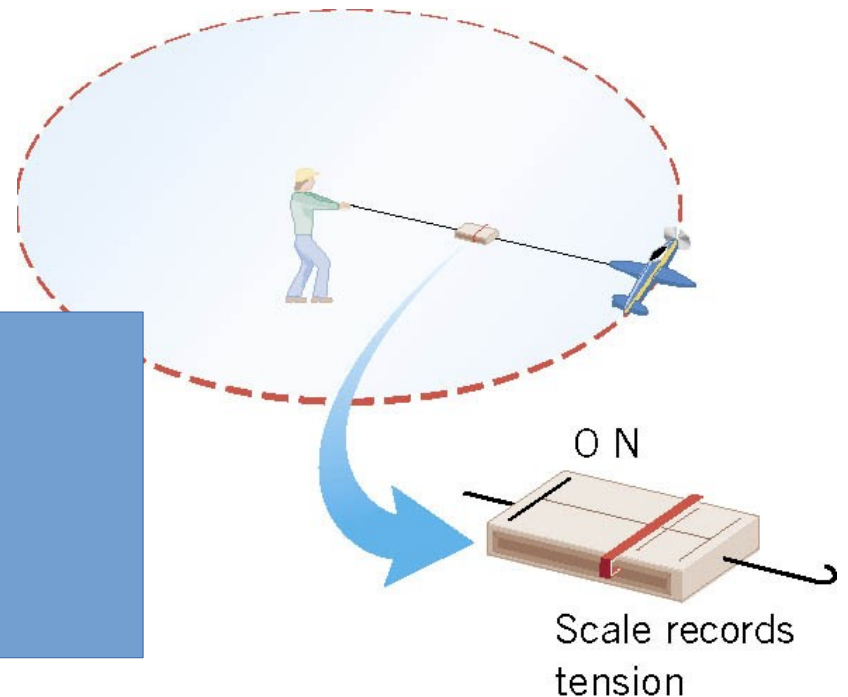
$$F_c = ma_c = m \frac{v^2}{r}$$

5.3 Centripetal Force

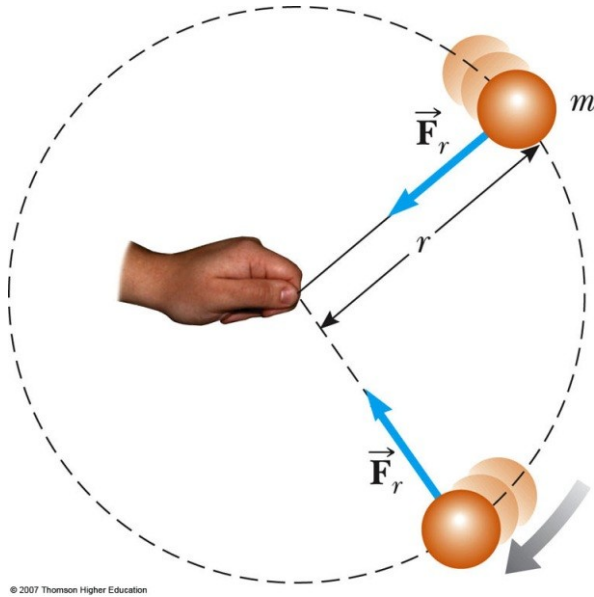
Example 5: The Effect of Speed on Centripetal Force

The model airplane has a mass of 0.90 kg and moves at constant speed on a circle that is parallel to the ground. The path of the airplane and the guideline lie in the same horizontal plane because the weight of the plane is balanced by the lift generated by its wings. Find the tension in the 17 m guideline for a speed of 19 m/s.

$$F_c = T = m \frac{v^2}{r}$$



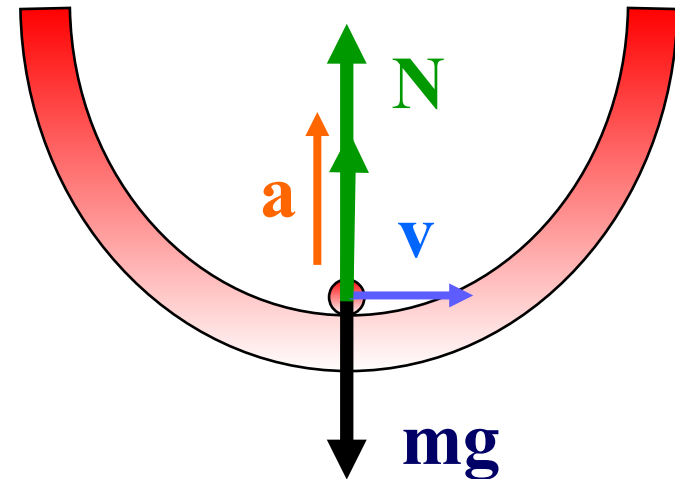
What provides Centripetal Force ?



$$F_{net} = T = ma$$
$$T = \frac{mv^2}{r}$$

$$F_{net} = N - mg = ma$$

$$N = mg + m \frac{v^2}{r}$$



5.3 Centripetal Force

Conceptual Example 6: A Trapeze Act

In a circus, a man hangs upside down from a trapeze, legs bent over and arms downward, holding his partner. Is it harder for the man to hold his partner when the partner hangs straight down and is stationary or when the partner is swinging through the straight-down position?

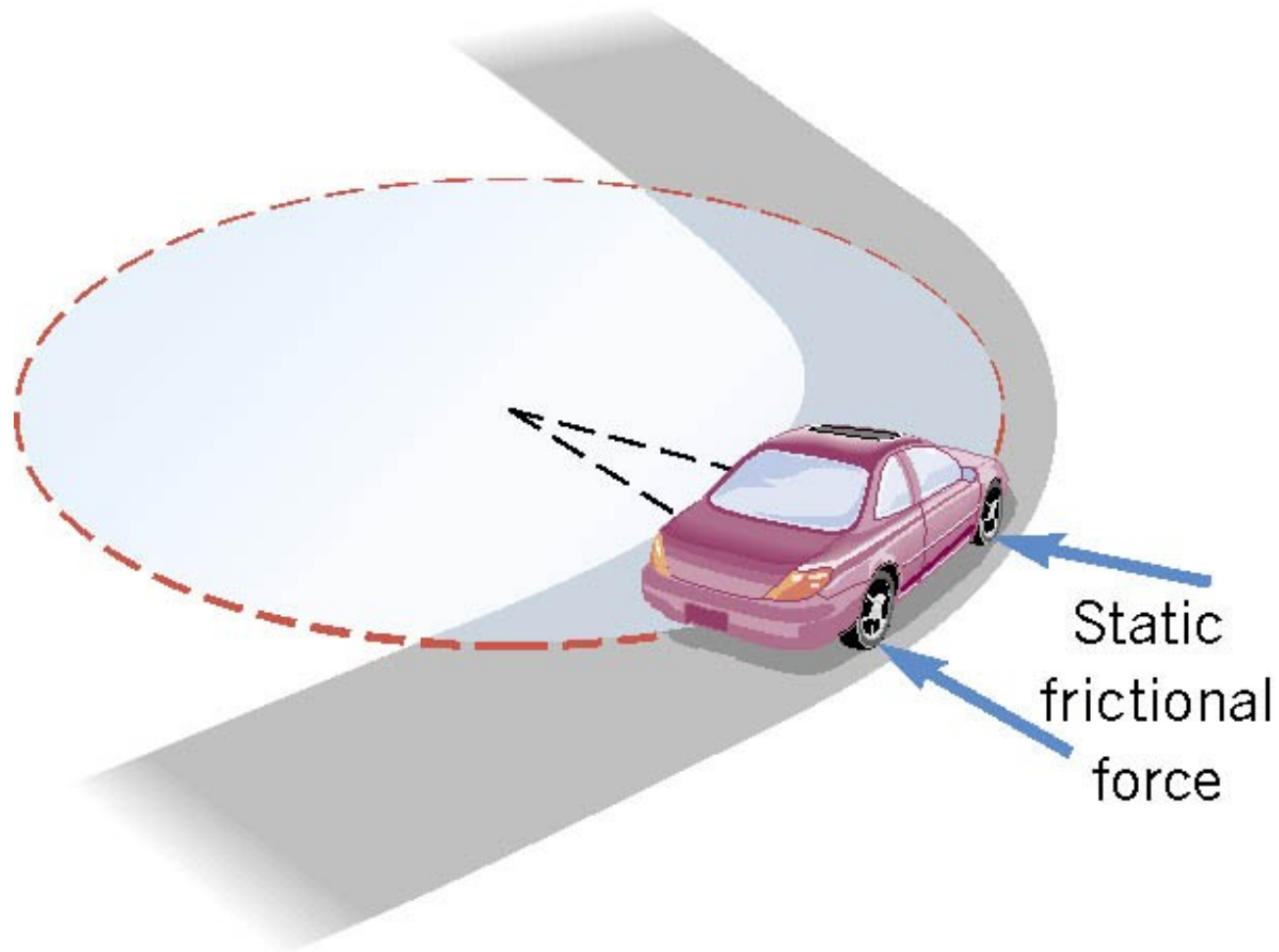


Discuss the free-body diagram.

In – out = mass $\times$ acceleration.

5.4 Banked Curves

On an unbanked curve, the static frictional force provides the centripetal force.



Revisiting uniform circular motion

In a previous unit we saw that the centripetal acceleration is v^2/R . v is the speed of the object moving in a uniform circular motion. If there is an acceleration $\rightarrow$ centripetal force that keeps the object Along the path.

Newton's second law: $F = ma \rightarrow \mathbf{F = m v^2/R}$

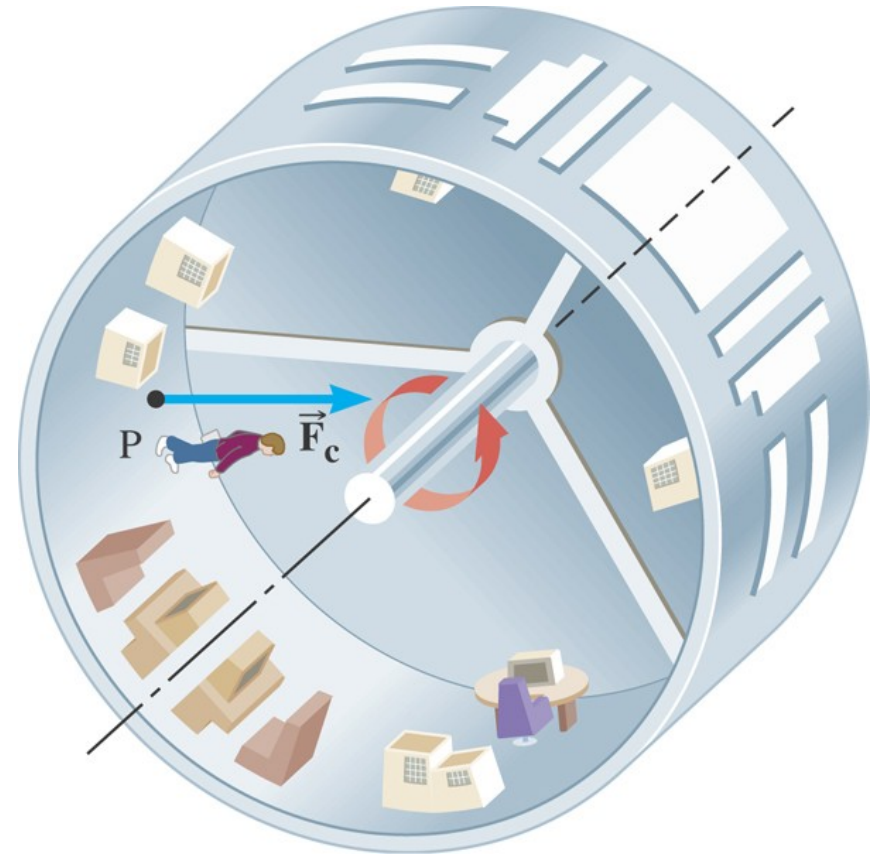
- 1) A 2kg rock is tied to a string 50cm long and spun around in a horizontal circle. If the string will break when the tension exceeds 250N , what is the greatest speed The rock can have without breaking the string ?
- 2) A bicycle and rider, with a combined mass of 50kg, are traveling at 5m/s. If they Are going in a circular path whose radius is 30m, how much force is the force Acting on the tired? (static frictional force).
- 3) The centripetal force due to gravity keeps satellite in orbit. The force is mass x g. g is 9.8 on Earth but g decreases with altitude. Suppose a satellite is at an Altitude of 12,000km (so radius $R = 12,000 + \text{radius Earth} = 18,400 \text{ km}$ about). At that altitude we have $g = 1\text{m/s/s}$ about.
Use $F = mg = \text{centripetal force } (\mathbf{F = m v^2/R})$ to find the orbital speed of the satellite
Hint: the mass are crossed out. Extract the speed v .

5.6 Apparent Weightlessness and Artificial Gravity

Example 13: Artificial Gravity

At what speed must the surface of the space station move so that the astronaut experiences a push on his feet equal to his weight on earth? The radius is 1700 m.

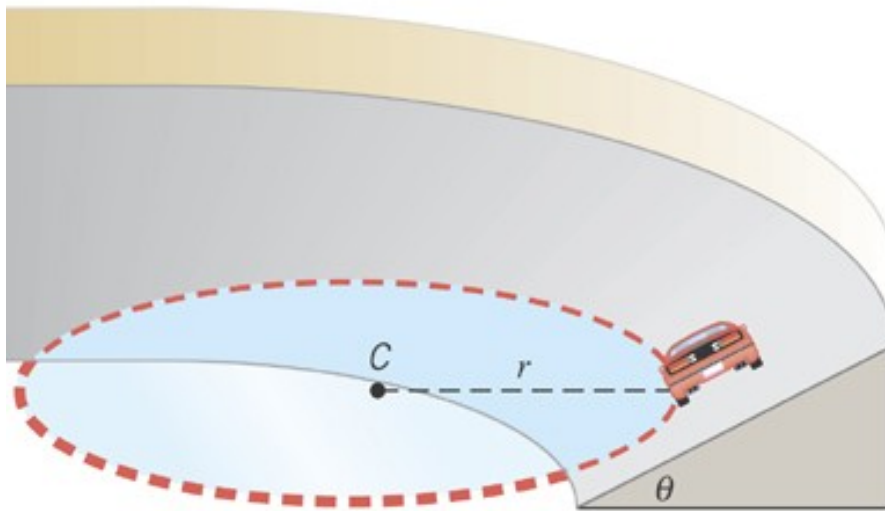
$$F_c = m \frac{v^2}{r} = mg$$



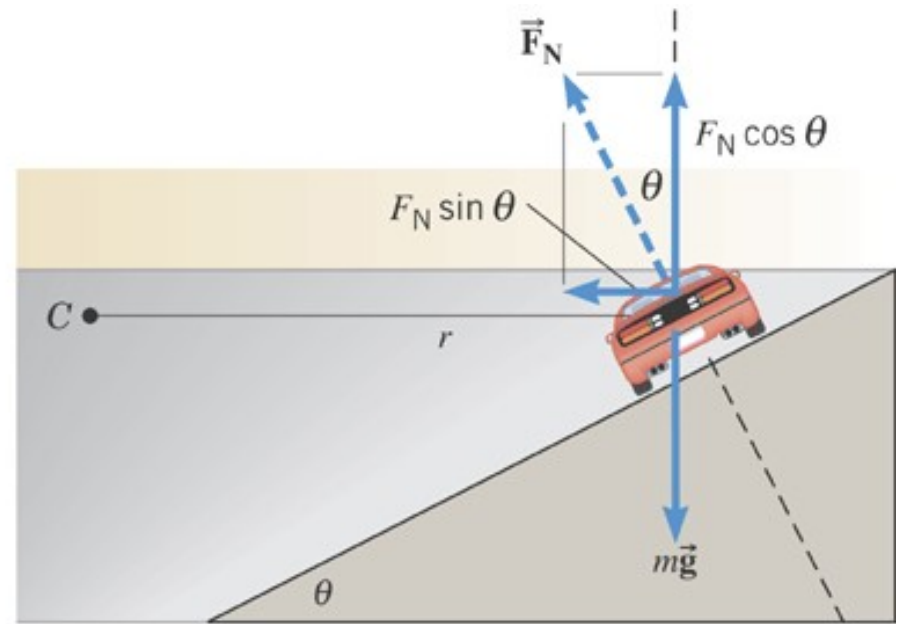
Space odyssey movie

5.4 Banked Curves

On a frictionless banked curve, the centripetal force is the horizontal component of the normal force. The vertical component of the normal force balances the car's weight.

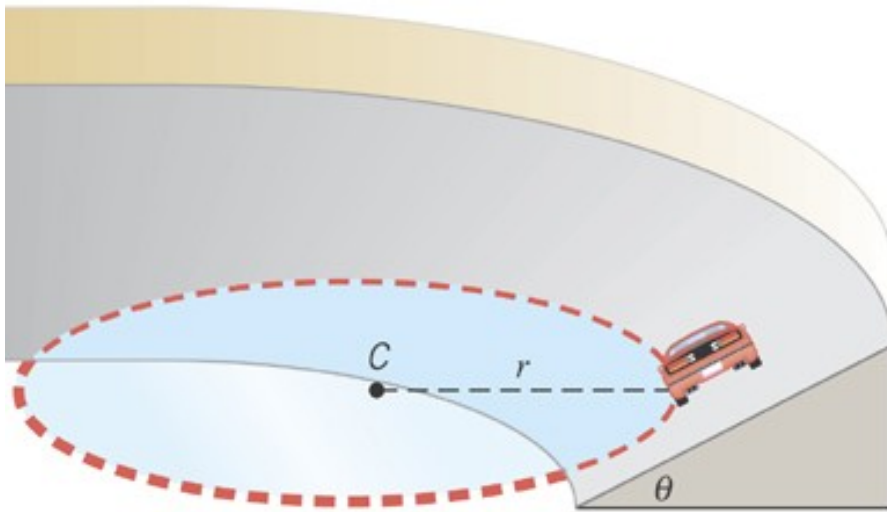


(a)

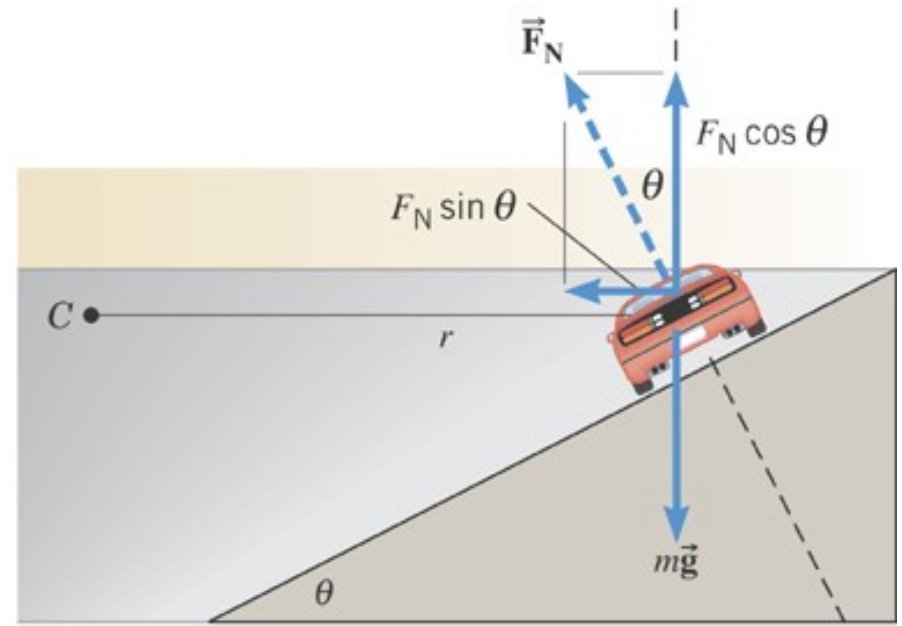


(b)

5.4 Banked Curves



(a)



(b)

$$F_c = F_N \sin \theta = m \frac{v^2}{r}$$

$$F_N \cos \theta = mg$$

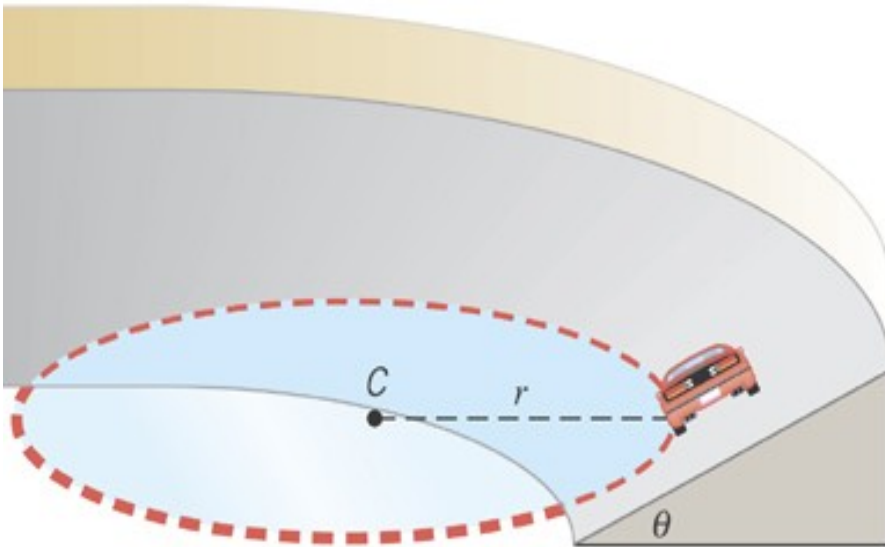
5.4 Banked Curves

$$F_N \sin \theta = m \frac{v^2}{r}$$

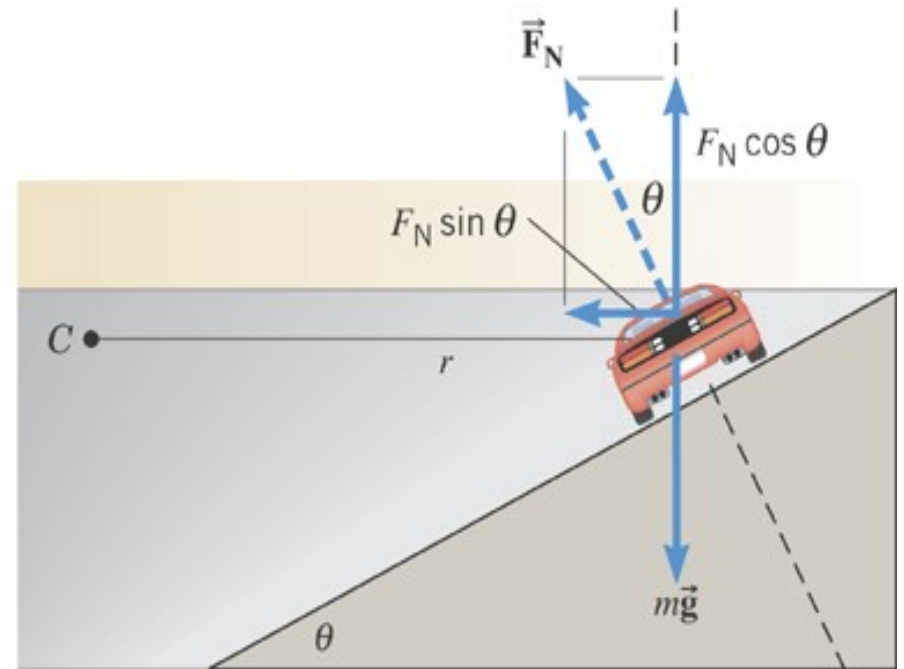


$$\tan \theta = \frac{v^2}{rg}$$

$$F_N \cos \theta = mg$$



(a)



(b)

5.4 Banked Curves

Example 8: The Daytona 500

The turns at the Daytona International Speedway have a maximum radius of 316 m and are steely banked at 31 degrees. Suppose these turns were frictionless. As what speed would the cars have to travel around them?



5.4 Banked Curves

Example 8: The Daytona 500

The turns at the Daytona International Speedway have a maximum radius of 316 m and are steely banked at 31 degrees. Suppose these turns were frictionless. As what speed would the cars have to travel around them?

$$\tan \theta = \frac{v^2}{rg} \quad \longrightarrow \quad v = \sqrt{rg \tan \theta}$$

$$v = \sqrt{(316 \text{ m})(9.8 \text{ m/s}^2) \tan 31^\circ} = 43 \text{ m/s} \quad (96 \text{ mph})$$



5.7 Vertical Circular Motion

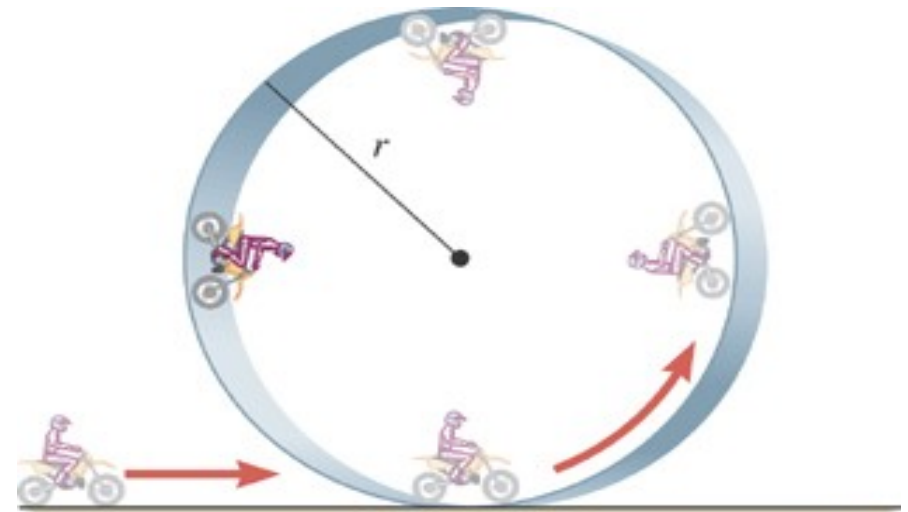
2

At the top $\rightarrow$ Gravity and the Normal force are pushing the biker in to keep it on track.

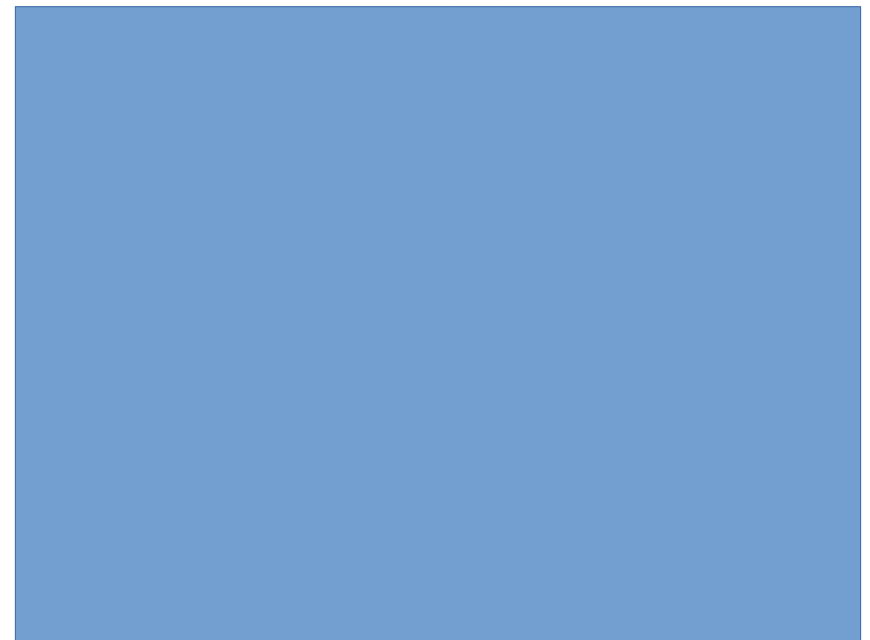
If $N = 0$, there is no contact $\rightarrow$ fall .

Set $N = 0$ to find the minimum Speed to make it over the loop.

Note : Newton's 2nd law $\rightarrow$
In - out = $m a$
For each point



(a)



(b)

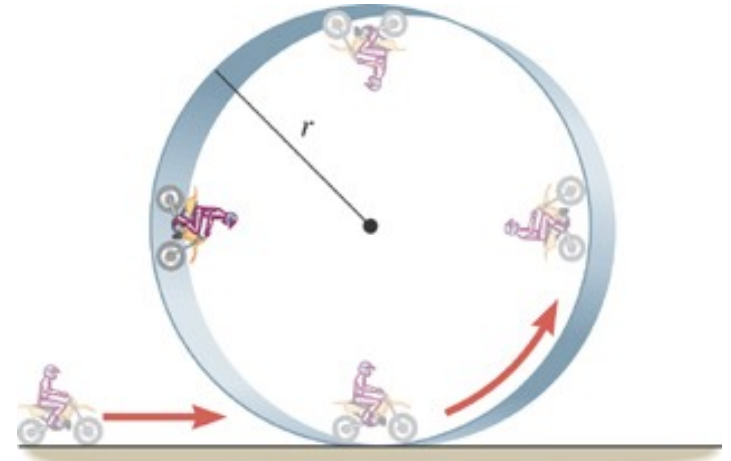
5.7 Vertical Circular Motion

$$F_{N1} - mg = m \frac{v_1^2}{r}$$

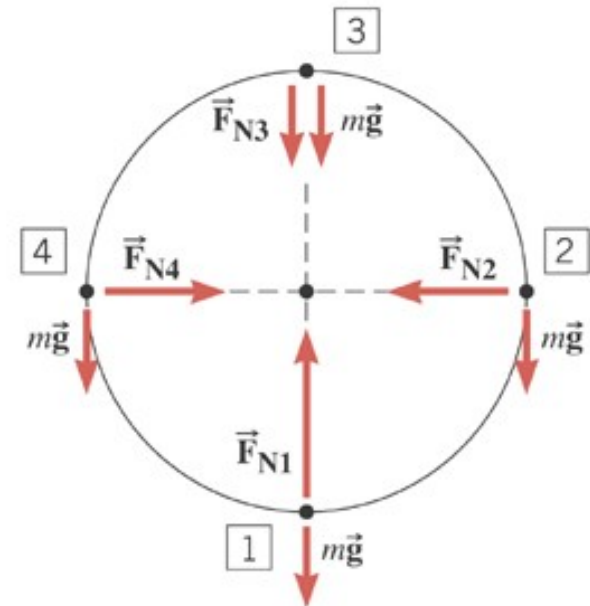
$$F_{N2} = m \frac{v_2^2}{r}$$

$$F_{N4} = m \frac{v_4^2}{r}$$

$$F_{N3} + mg = m \frac{v_3^2}{r}$$

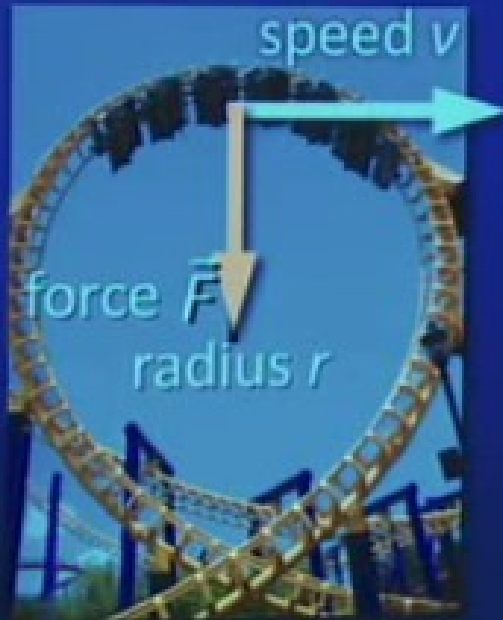


(a)



(b)

Loop-the-Loop: How fast?



Doing the numbers:

“The Revolution,”
Six Flags Magic Mountain,
Valencia, California

$r = 6.3$ meters

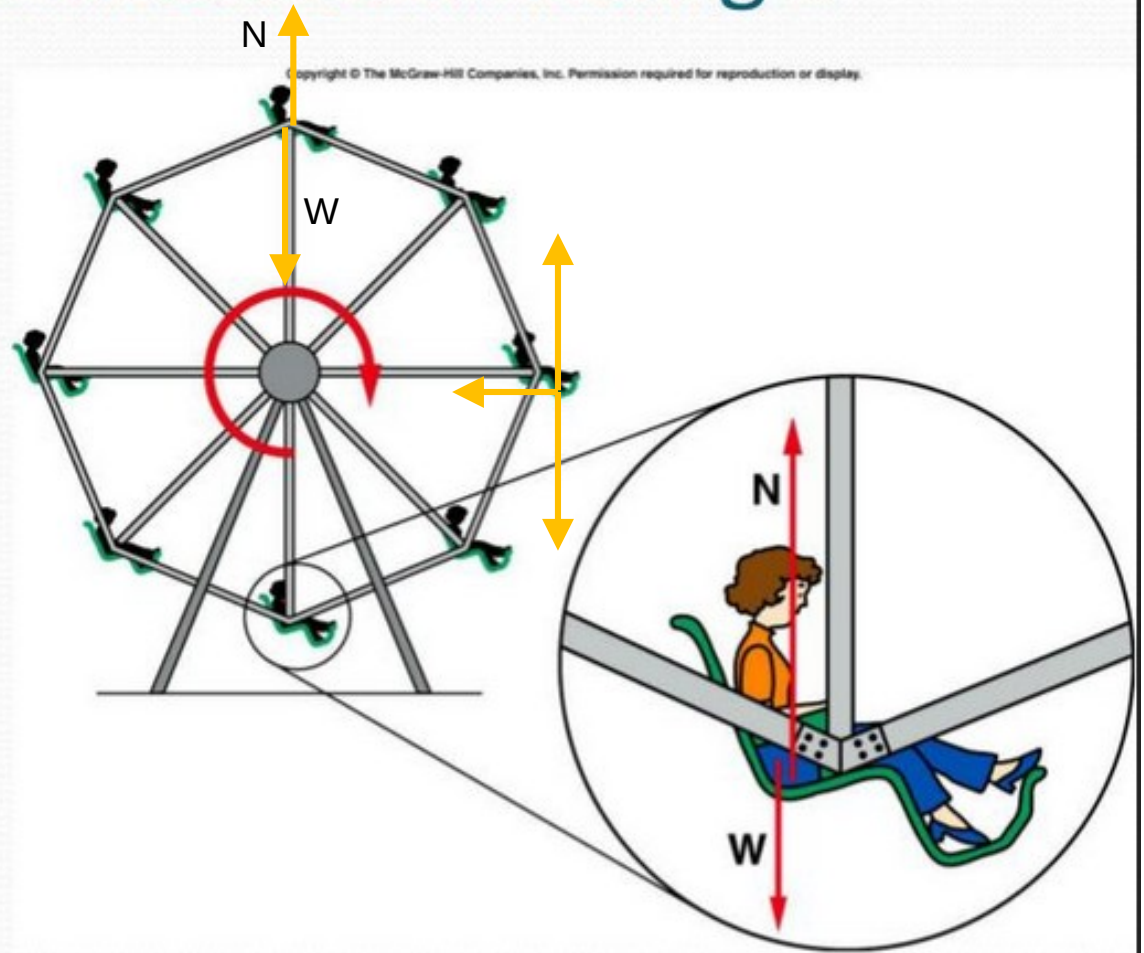
$$v = \sqrt{gr} = \sqrt{(9.8 \text{ m/s}^2)(6.3 \text{ m})} = 7.9 \text{ m/s} \quad (18 \text{ miles/hour})$$

Here again. At the top → **In – out = m x acceleration**

What forces are involved in riding a Ferris wheel?

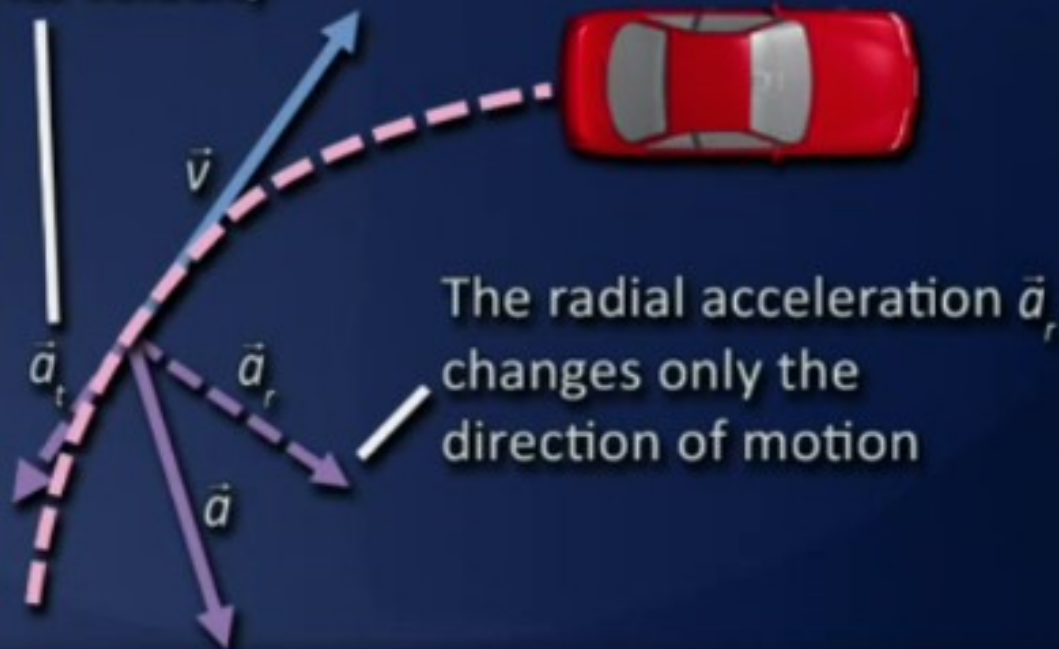
Depending on the position:

- **Weight of the rider**
- **Normal force from seat**
- **Seat-belt force**

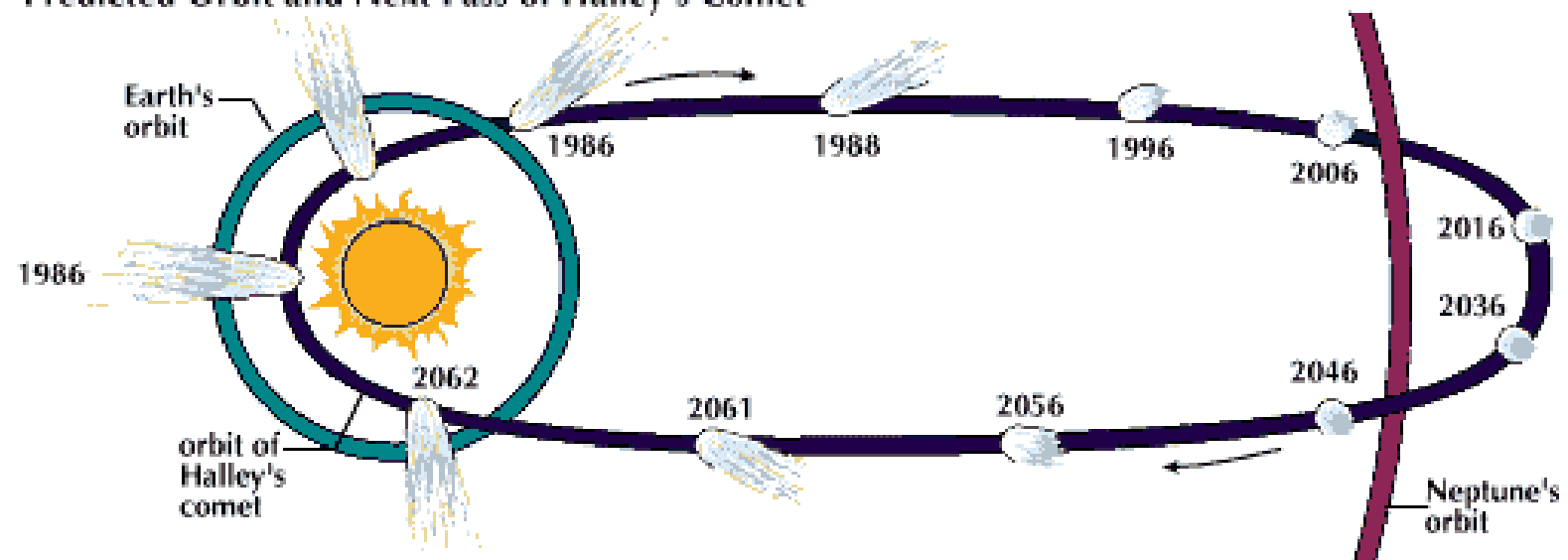


Nonuniform Circular Motion

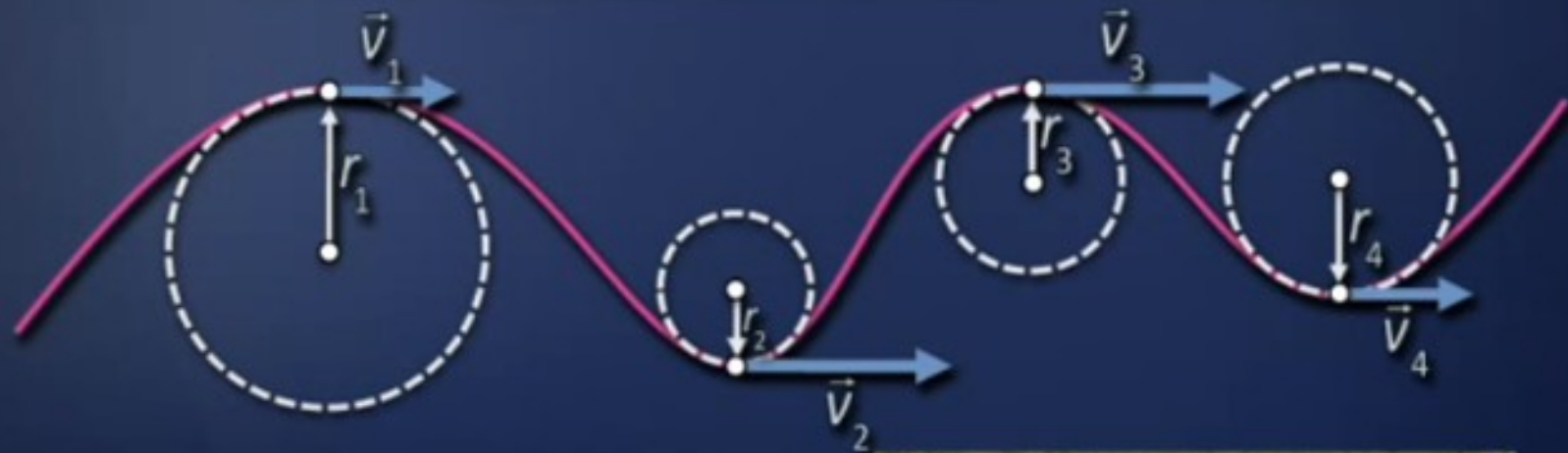
The car is slowing, so its tangential acceleration $\vec{a}_t$ is opposite its velocity



Predicted Orbit and Next Pass of Halley's Comet



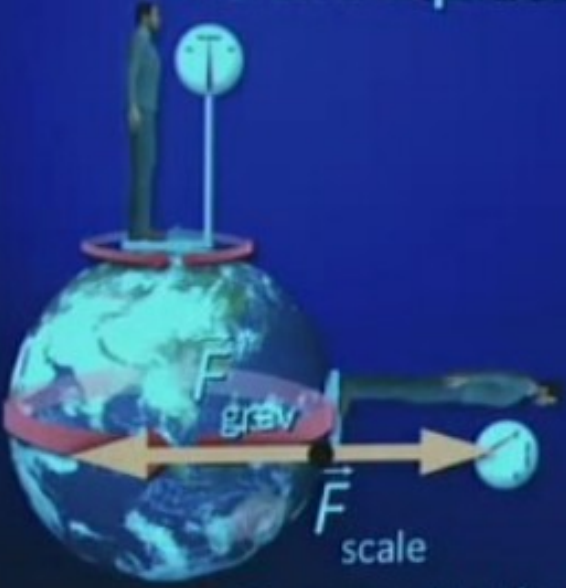
Nonuniform Circular Motion



Osculating circle



At the Equator: What's the scale read?



Acceleration direction?
Toward Earth's center

So net force: toward center

Forces acting?
Gravity (down)
Scale (up)

So gravitational force (actual weight)
is larger than scale force

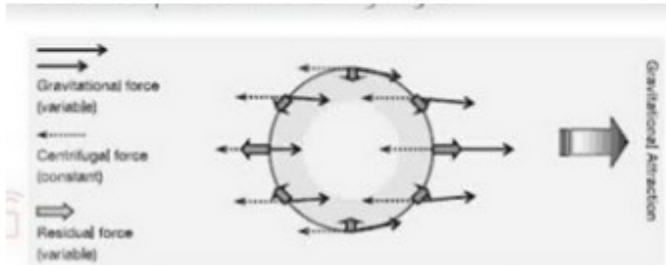
Result: Apparent weight less than actual weight

$In - out = ma.$

Your apparent weight is the normal force $<$ weight at equator.

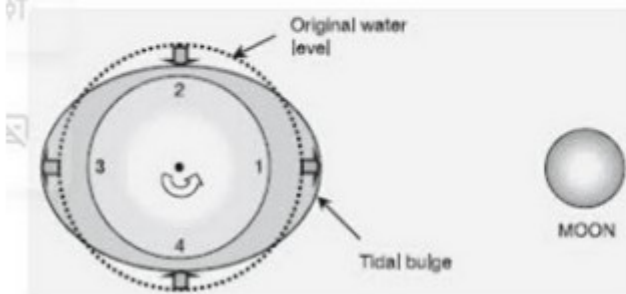
Tides: Differential forces from the Moon (mostly)

- The force by the Moon on the Earth is **greatest** on the **near** side and **smallest** on the **far** side.
- This causes two high tides at each of these points.
- There are **TWO** high tides per 24 hours!

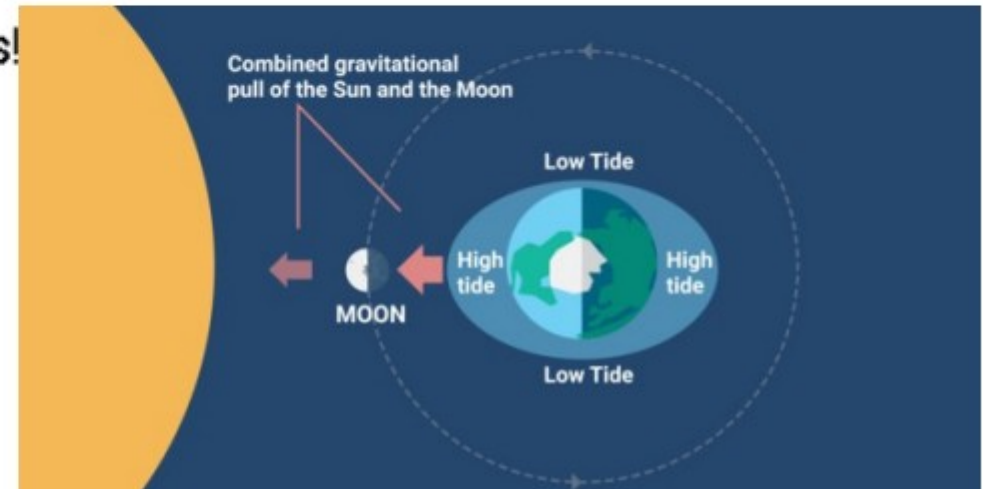


If the Earth were completely covered by a very deep ocean this residual force would act on the water and produce two bulges, one on the side facing the Moon and one on the opposite side of the Earth.

The following diagram shows the Earth and Moon as viewed from above the North Pole. An observer under the water on the Earth at position 1 is beneath one of the water bulges and would be experiencing a high tide. A quarter of a revolution later, at position 2 where the original water level has been depressed, a low tide would be seen. The high tide, low tide sequence repeats as the observer moves around through points 3 and 4.



The Earth, Sun, and Moon are subject to similar gravitational and



Spring tides.

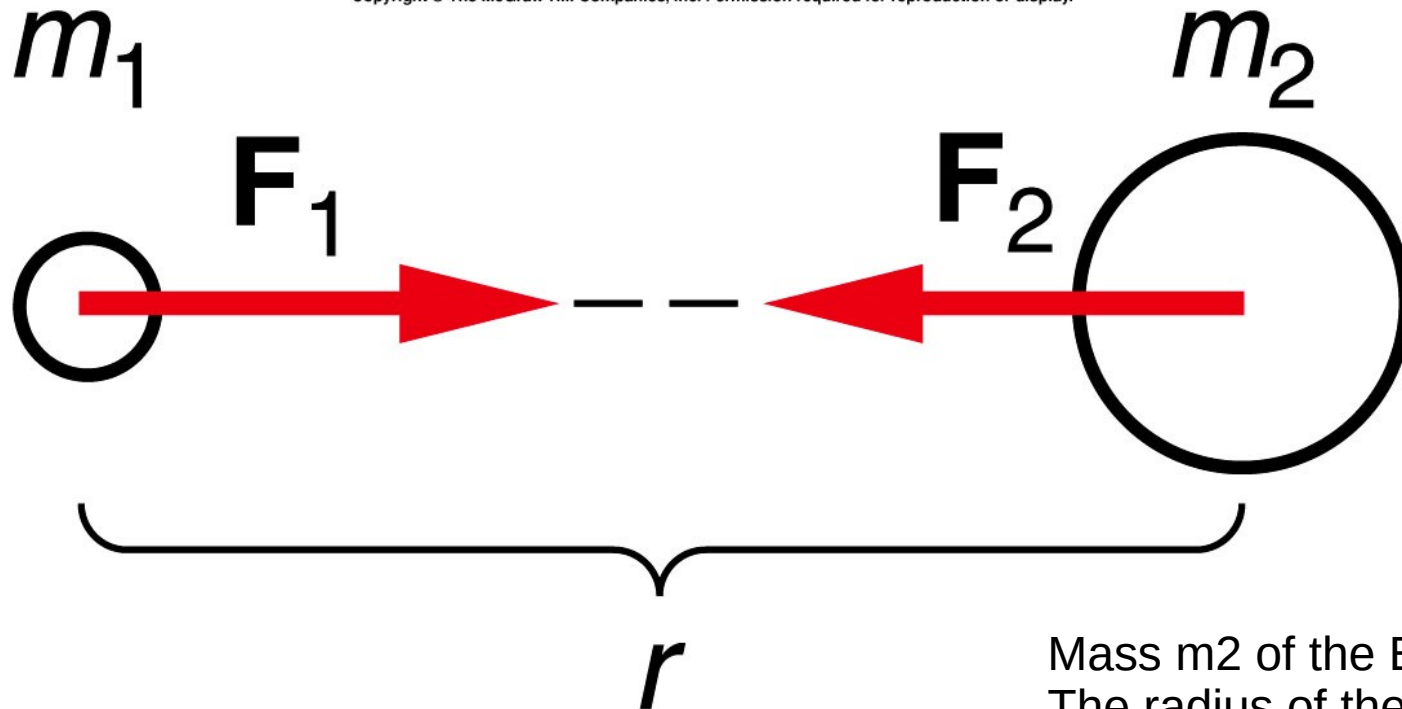
These more extreme tides, with a greater tidal range, are called spring tides. Spring tides don't just occur in the spring; they occur whenever the Moon is in a new-moon or full-moon phase, about every 14 days. Spring tides occur when the tidal bulges from the Moon and Sun are aligned.

neap tide The tide of small range that occurs every 14 days, near the times of the first and last quarter of the Moon, when the Moon, Earth, and Sun are at right angles.

Let's revisit weight = force due to gravity and newton's third law.

Newton unified weight, motion on earth and motion in the heaven
With his universal law of gravitation.

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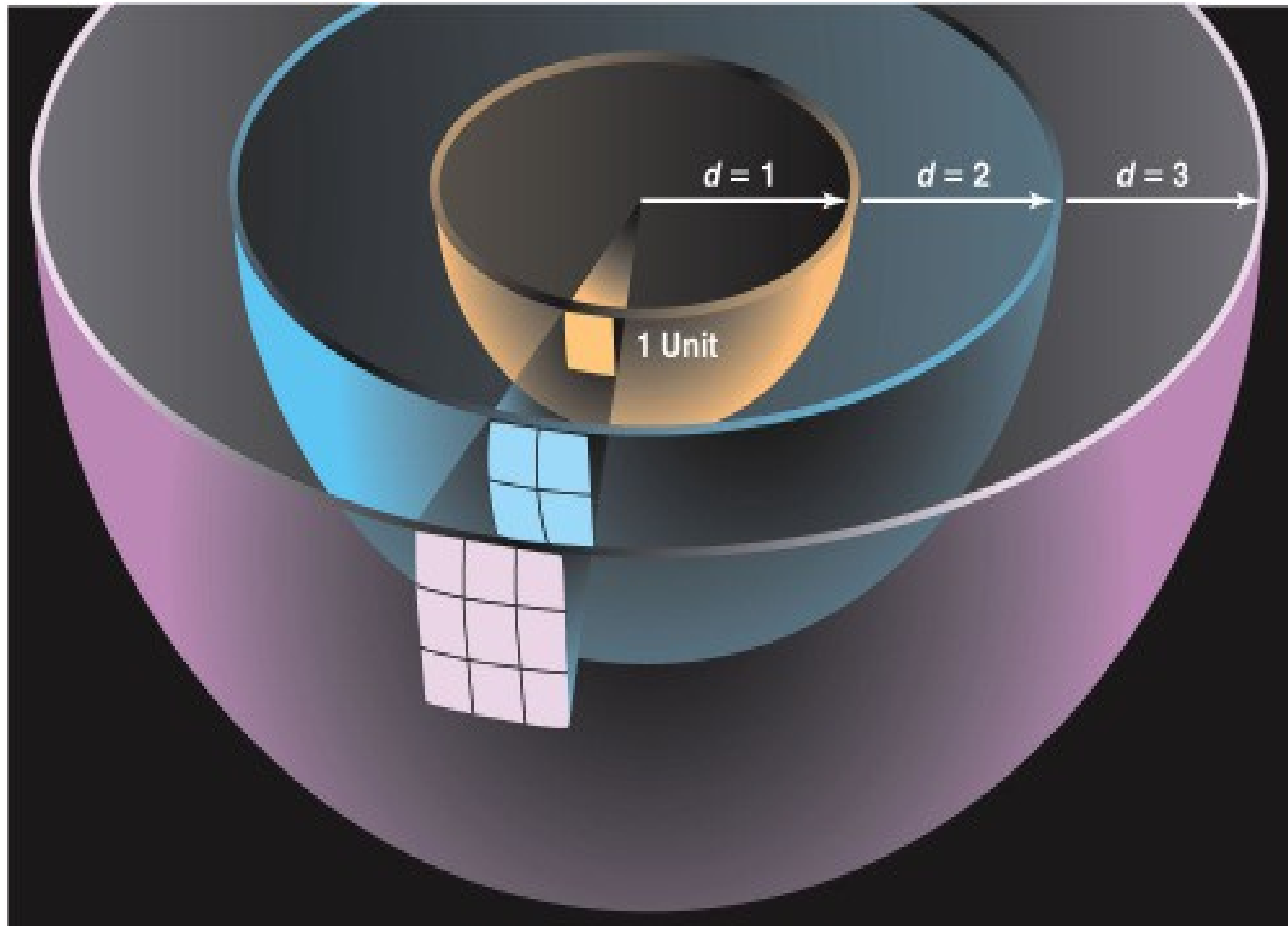
$$G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$$

$$F = G \frac{m_1 m_2}{r^2}$$

Mass m_2 of the Earth is $5.972 \times 10^{24} \text{ kg}$
The radius of the Earth is 6,371 km
If you mass m_1 is 60kg:
A) find the pull due to gravity
B) compare to your weight.

Textbook 6.6 and 6.7

Gravity is an inverse square law – lab inverse square law

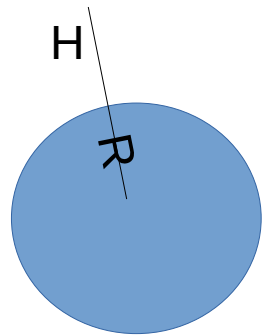


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If you were to compress the Earth to the size of the Moon. The factor is 4.
What will happen to your weight?

See exploration of physical sciences. gravity

Supposed the earth is a sphere of radius $R = 6,400$ km about



At the surface $g = 9.8$ m/s/s
You weight is $mg = G m M / R^2$
So $g = G M / R^2$

Likewise at the altitude H $g' = G M / (R+H)^2$
So $g'/g = R(R+H)^2$
Or **$g'/9.8 = R(R+H)^2$**

- 1) Find the acceleration due to gravity g' for $H = 12,000$ km
- 2) find the acceleration of a piece of junk at an altitude of 22,000km.

Movies:

- sad and happy ball
- Astronaut and third law